

1 Empirical Application

1.1 Data Description

We use data from ‘*dunnhumby*’, a customer data science company based in the UK. The specific dataset we use is named ‘*The Complete Journey*’ and comprises household level transactions spanning two years (March 2016 to March 2018) from a group of 2,490 households who are frequent shoppers at a retailer. We observe several household attributes (age, income bands, home-ownership, household size, marital status) as well as in-store product placement (other than regular product placement e.g., store front, store rear, etc.).

As is common in the literature (Echenique, Lee, and Shum, 2011; Dean and Martin, 2016), we focus on the (ubiquitously assumed to be) income-separable group of food and beverages. For each unique product, we observe a specific description of the product (e.g., salmon) as well as a higher level of grouping (e.g., seafood). Based on this detailed product-level information, we group goods into 9 categories from which we observe 1,092,840 transactions from 28,302 unique products.¹ We aggregate purchases by fortnight to minimize the effect of goods more prone to being storable whilst maximizing our ability to detect any dynamics in rationality. As the number of transactions in March - June 2016 and March 2018 are substantially lower than the other time periods, we drop these observations leaving us with 41 time periods.

As is also common, we assume prices are the same for each household as not all goods are bought in all observations across all the households. We construct monthly price indices for each of the categories by a weighted average. Specifically, for each category, let $\mathcal{C}$ denote all products in that category. The monthly price index $P_{\mathcal{C},t} = \sum_{c \in \mathcal{C}} \omega_{c,t} \times p_{c,t}$, where $p_{c,t}$ represents the mean price per unique product, and $\omega_{c,t}$ is the budget share for each unique product across all households. A household’s consumption is defined as their total expenditure for each category and time period divided by the respective price index. Dropping any households with no consumption in any of the categories, we are left with 2474 households. See Online Appendix, Section B.1 for further details.

1.2 Are households badly-behaved?

We compute a ‘regular’ set of indices that are commonly used in the literature, as well as their ‘consumer’ analogues. For the regular indices, we focus on the Afriat Efficiency Index (AEI) (Afriat, 1967), the Average Varian Index (AVI) (Varian, 1990),

¹Categories: dairy, fruits and vegetables, meat and seafood, non-sugary beverages, prepared foods, savory snacks, starches, sugary beverages, sweet snacks.

the Houtman-Maks Index (HMI) (Houtman and Maks, 1985), and the Money Pump Index (MPI) (Echenique et al., 2011).² The consumer indices we compute are what we refer to as the C(onsumer)-AEI, C-AVI, C-HMI, C-MPI. Although Apesteguia and Ballester (2015) recommend an extension to their Swaps Index (SI) for the infinite-dimensional case (as opposed to finite choice) in Appendix D of their paper, a rigorous methodology and implementation is yet to be found in the literature, hence estimation is omitted. That being said, we compute an analogous CS index by approximating the per-period Lebesgue measure using Monte-Carlo simulation.

A well-known issue with rationality indices is their predictive power, i.e. the ability to truly detect irrational behaviour. For example, if budget sets rarely overlap, even random choices will generate few violations and, as an artifact, any reasonable index will measure as essentially rational. To address this, we follow Dean and Martin (2016) and compute two predictive power measures for each index. The first is the *Selten score*, defined as a household’s rationality index minus the average of rationality indices obtained from simulated choices made when facing identical budget sets. Thus, given any index that measures perfect rationality as 0, a negative/positive Selten score indicates the household behaves more/less rationally than random choice. The second measure is the *percentile*, which uses the distribution of random choosers’ rationality from which we calculate the percentage of simulated random choosers whose rationality index is at least as high as the household’s rationality index. For example, a household at the x^{th} percentile is more rational than $x\%$ of simulated households facing the same budget sets.

A key point to discuss in using the Selten score and percentile is how to simulate the random choices. Although it would be intuitive to construct random choices by drawing uniformly randomly over budgets to obtain simulated household datasets, a more robust approach would involve using observed budget choices. Even with a moderate number of categories, households may rarely purchase from each category across periods. This implies that households are often choosing from “corners” of budget sets. As uniform draws are rarely sampled from these corners, such draws would not serve as a suitable benchmark; datasets drawn uniformly may appear highly rational as it would be increasingly difficult to generate violations with a larger number of categories.³ To

²AVI and HMI are computed using the mixed-integer linear programming method from Demuynck and Rehbeck (2023). As the MPI is a known NP-hard problem, Smeulders, Cherchye, Spieksma, and Rock (2013) propose and implement an alternative algorithm for the minimum and maximum MPI; we take the maximum as the measure of MPI.

³This is in contrast to Dean and Martin (2016) in which the authors focus on uniform draws, which is more appropriate given their main analysis involves just 3 categories. They also perform a robustness check where they draw from the empirical distribution of budget shares combined across all households, instead of a theoretical distribution based on the budget shares as we do. Our approach generates a well-defined and plausible stochastic benchmark rather than depending on uniform draws or other

address this, we simulate 100 random choice datasets for each household by drawing from household-specific Dirichlet distributions where the concentration parameters are determined by the observed per-period budget shares.⁴

Table 1 summarizes the estimation of the different rationality indices with $1748/2474 = 70.1\%$ of households violating GARP/SARP. For example, with average expenditure (across household and time) being \$35.55, an average (1-)C-AEI of 9.93% means households could have maximally saved approximately \$3.53 per period on average by buying cheaper bundles whilst attaining the same utility. Negative Selten scores and high percentiles across the board suggest that the indices have sufficient predictive power. Naturally, we should expect a large degree of heterogeneity which we can observe from Figure 1. By comparing the histograms which use household data against the simulated data, households are more rational than random choosers, again suggesting that our modeling of rationality is sufficiently powerful.

From Table 2, the indices are all positively correlated with each other which is natural given all measures are tasked with the same objective of essentially quantifying the “degree” of cyclicity in the data. It is clear from Table 2 that although positive correlations exist, there are no pairs of perfectly correlated indices implying that all measures are (to some non-negligible degree) independent in their measurement of irrationality.

1.3 What types of households are more rational?

Tables A1 and A2 contain regression results for the rationality indices on the available demographic variables: age, income, home-ownership, household size, and marital status.⁵ We restrict this regression analysis to the 801 households for which we observe complete demographic data. Looking at the consumer indices, generally, households that have household heads who are older, have lower incomes, or live in larger households are less rational. Conversely, households where the head is single or a homeowner

households.

⁴We model each household’s purchasing patterns using a household-specific Dirichlet distribution estimated via Maximum Likelihood Estimation (MLE). First, we calculate empirical budget shares for each household at every time period (removing any goods that are never purchased across the entire sample). For each household, using their full collection of observed budget shares, we estimate good-specific concentration parameters (α_k) and an overall scale parameter (α_0 , where $\alpha_0 = \sum \alpha_k$). The relative sizes of the α_k parameters reflect the household’s average budget allocation for each good, while α_0 dictates the consistency of their spending, where a higher value implies stable budget shares across periods and a lower value indicates more volatile, dispersed budget shares. After mapping zeros back to the omitted goods, we use these fitted parameters to draw random budget share vectors from the resulting Dirichlet distribution for each period, which are scaled by actual total expenditures and divided by prices to reconstruct the counterfactual random quantities.

⁵The excluded categories are the youngest, lowest-income, unknown home-ownership status, smallest household size, and unknown marital status. Also, unfortunately, we do not observe any data on educational attainment.

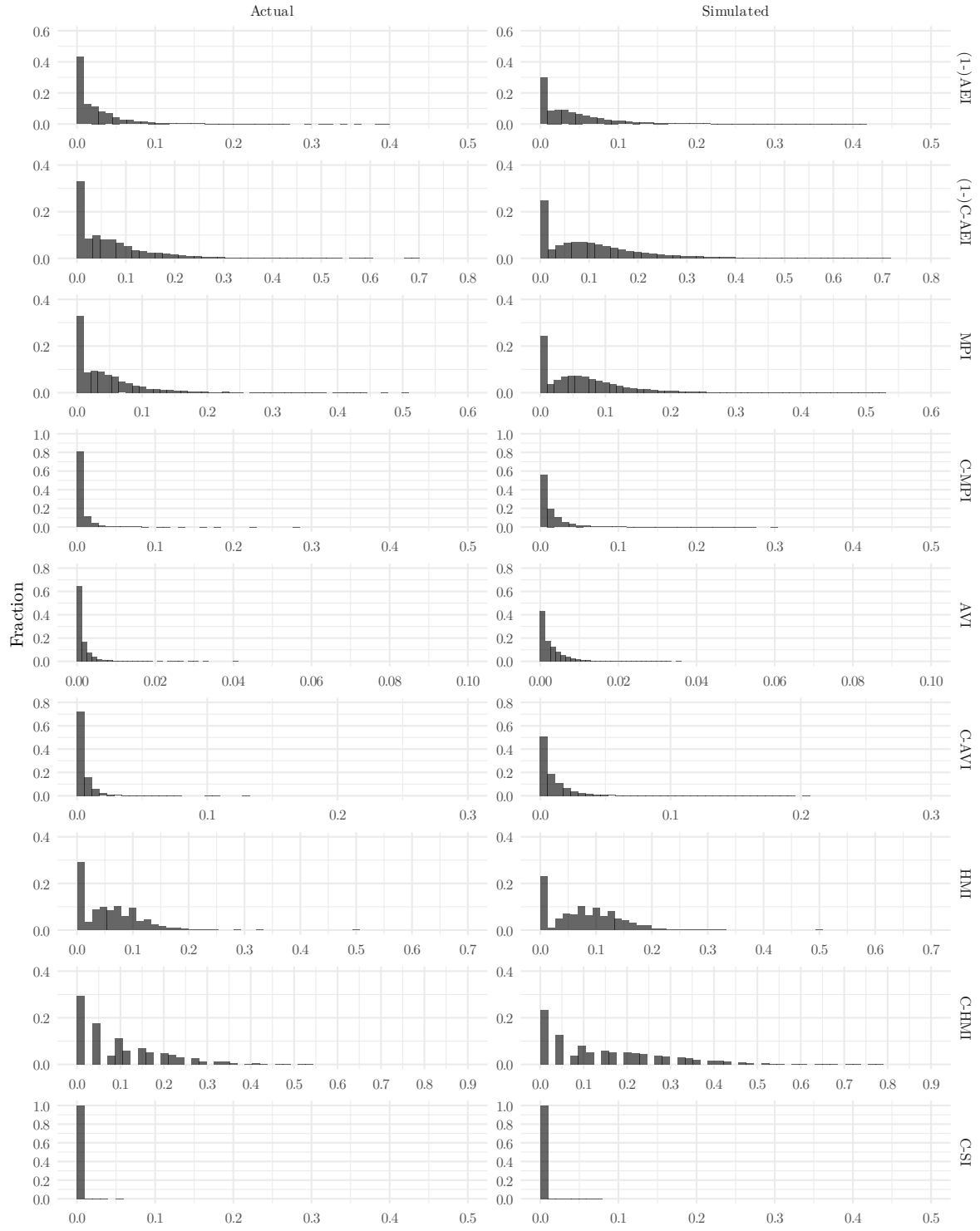


Figure 1: Histograms of regular and consumer indices from household data and simulated data.

		Mean (%)	Selten score (%)	Percentile (%)
Consumer	(1 -)AEI	9.93	-3.66	73.28
		(8.92)	(8.39)	(30.98)
	MPI	0.88	-0.78	76.87
		(1.56)	(1.5)	(28.24)
	AVI	0.73	-0.56	75.43
		(1.04)	(1.03)	(29.59)
	HMI	14.18	-5.05	75.15
		(8.89)	(8.13)	(28.44)
	SI	0.1	0	92.48
		(0.49)	(0.15)	(25.13)
Regular	(1 -)AEI	3.83	-1.62	72.73
		(4.56)	(4.14)	(30.9)
	(A)MPI	6.5	-2.29	73.3
		(6)	(5.58)	(31.12)
	AVI	0.22	-0.13	74.58
		(0.32)	(0.29)	(29.48)
	HMI	8.03	-2.13	79.27
		(4.29)	(4.72)	(26.05)

Table 1: Summary statistics of rationality indices for badly-behaved households. Standard deviations shown in parentheses.

exhibit more rational behavior. While lacking statistical significance, the estimated effects for age and income mirror those in Echenique et al. (2011) and Dean and Martin (2016), whereas the results flip direction for household size.

The equivalent regressions with the Selten scores are presented in Tables A3 and A4.⁶ Focusing on the consumer indices, firstly, similar to Dean and Martin (2016), using the Selten score yields more statistically significant coefficients. Secondly, we still observe that household heads who are married or older remain less rational overall. Notably, income undergoes a clear directional reversal with middle- and upper-income cohorts shifting from being more rational to less rational in the Selten score regressions. This reversal aligns with the intuition in Aguiar and Hurst (2007) that households facing a higher opportunity cost of time have less incentive to optimize their purchases. As a result, random behavior that may have looked rational in raw terms is revealed as less rational once the Selten score is used.

⁶Dean and Martin (2016) argue that the rationality indices may not account for unobserved heterogeneity among certain clusters of households, e.g., number of overlapping budget sets, making a reasonable case for using the Selten score.

	(1-)C-AEI	C-MPI	C-AVI	C-HMI	(1-)AEI	MPI	AVI	HMI
(1-)C-AEI	—	0.703	0.848	0.567	<i>0.831</i>	0.975	0.798	0.536
C-MPI	0.674	—	0.782	0.454	0.647	<i>0.675</i>	0.755	0.429
C-AVI	0.829	0.759	—	0.641	0.793	0.826	<i>0.766</i>	0.470
C-HMI	0.353	0.363	0.577	—	0.494	0.541	0.454	<i>0.792</i>
(1-)AEI	<i>0.791</i>	0.605	0.759	0.325	—	0.903	0.901	0.446
MPI	0.966	<i>0.639</i>	0.801	0.321	0.883	—	0.847	0.521
AVI	0.768	0.728	<i>0.732</i>	0.312	0.886	0.827	—	0.497
HMI	0.307	0.351	0.352	<i>0.663</i>	0.258	0.291	0.395	—

Table 2: Combination of correlation matrices for the rationality indices. The lower triangle displays correlations where zeros (perfect rationality) are excluded, with the upper triangle displaying correlations with zeros included. All estimates are statistically different from zero at any reasonable significance level.

1.4 Does rationality change over time?

As the AEI, AVI, and MPI are all notionally based on quantifying overall cost (in)efficiency, the C-AEI, C-AVI, and C-MPI are defined naturally as (different) aggregators over the per-period cost efficiencies as defined in [Section XXX](#). Unlike the regular indices, the existence of a well-defined cost efficiency for each time period allows us to investigate how rationality may vary over time; we call this measure the Cost Efficiency Index (CEI).⁷ For other indices which are methodologically different from the notion of cost (in)efficiency, such as the HMI (dropping bundles to satisfy SARP) and CS (alternatives that dominate the observed choice), an analogous investigation of rationality dynamics is also entirely possible; for brevity, we focus on the dynamics of the CEI (and the Selten score).

In Figure 2, which plots the household average over time, looking at the top two lines, we can clearly observe that rationality is not constant, although visually there does not appear to be any obvious upward or downward trends, with the CEI fluctuating around its means (0.051 excluding perfectly rational periods, and 0.005 including them, respectively). We do observe some distinct patterns when comparing 2016 and 2017;

⁷A potentially simple way of measuring rationality over time using the regular indices is to estimate indices for subsets of a dataset over time. However, this is a somewhat ad-hoc approach, lacking theoretical foundation, making it unappealing and hard to interpret properly.

some cyclical/seasonality emerges in mid-November and mid-December, when the CEI appears to substantially increase, suggesting that there is something about those times of the year that leads to households buying costlier bundles. These patterns remain whether we exclude or include the perfectly rational periods. De-measured OLS regression analyses with time dummies reported in Tables A7 and A8 confirm this cyclical pattern. We also include the same demographic variables as the previous static analysis and also a set of variables called display promos which include, for every time period, a household’s total purchase frequency of products featured in non-standard display locations. Additional estimates from random effects (RE) regressions (with controls) support that the cyclical pattern is generally robust to household heterogeneity.⁸

The third line of Figure 2, which plots the Selten score over time is similar in the sense that there is no obvious trend, with large downward deviations during the same time periods as the CEI had substantial increases. In contrast with the CEI, such deviations in the Selten score imply that either i) household rationality increases in those time periods, ii) simulated household rationality decreases, or iii) both. As we observe household rationality substantially decreasing in those periods, it must be that the simulated households are significantly more irrational, making households appear relatively more rational. As such, the Selten score provides powerful evidence that the increases in the observed household irrationality during those periods reflect genuine behavioral responses.⁹

Next, we run sets of OLS and RE regressions involving a simple time trend (with and without controls). Table A5 indicates that there is a positive time trend suggesting that rationality is marginally decreasing over time with Table A6 suggesting a slight upward trend relative to simulated households.¹⁰ Although a time trend is detected (relative to the mean) and is robust to controls, the estimated effect is small. A very natural question arises from this, does previous period(s) rationality help explain current period

⁸From the OLS estimates (with and without controls) using the CEI, we also perform a multiple hypothesis test comparing the coefficients from the same times of year but different years (if available) i.e. comparing the coefficients for the dummy variable for [2016-07-25 to 2016-08-07] with the dummy for [2017-07-24 to 2017-08-06], and so on. Using Bonferroni correction, we are unable to reject that the pairwise differences in the coefficients are different from 0 at any reasonable significance level, except for the dummies for [2016-12-12 to 2016-12-25] and [2017-12-11 to 2017-12-24] which are the time periods with the large jumps in irrationality. For the SS regressions, we fail to reject any of the hypotheses tests. Identical conclusions hold when using other corrections, e.g., Holm’s method, Benjamini & Hochberg’s false discovery rate method.

⁹For example, if the Selten score had jumped up to 0 (or more), then the observed jump in irrationality could just as easily be explained by households resorting to “more” random behaviour. The fact that the score drops instead indicates they are actually outperforming chance by a wider margin despite the more difficult budget sets.

¹⁰Hausman tests in favor of RE over FE at any reasonable level of significance. For both sets of regressions, LR-tests for OLS vs. RE also suggest that significant heterogeneity is present.

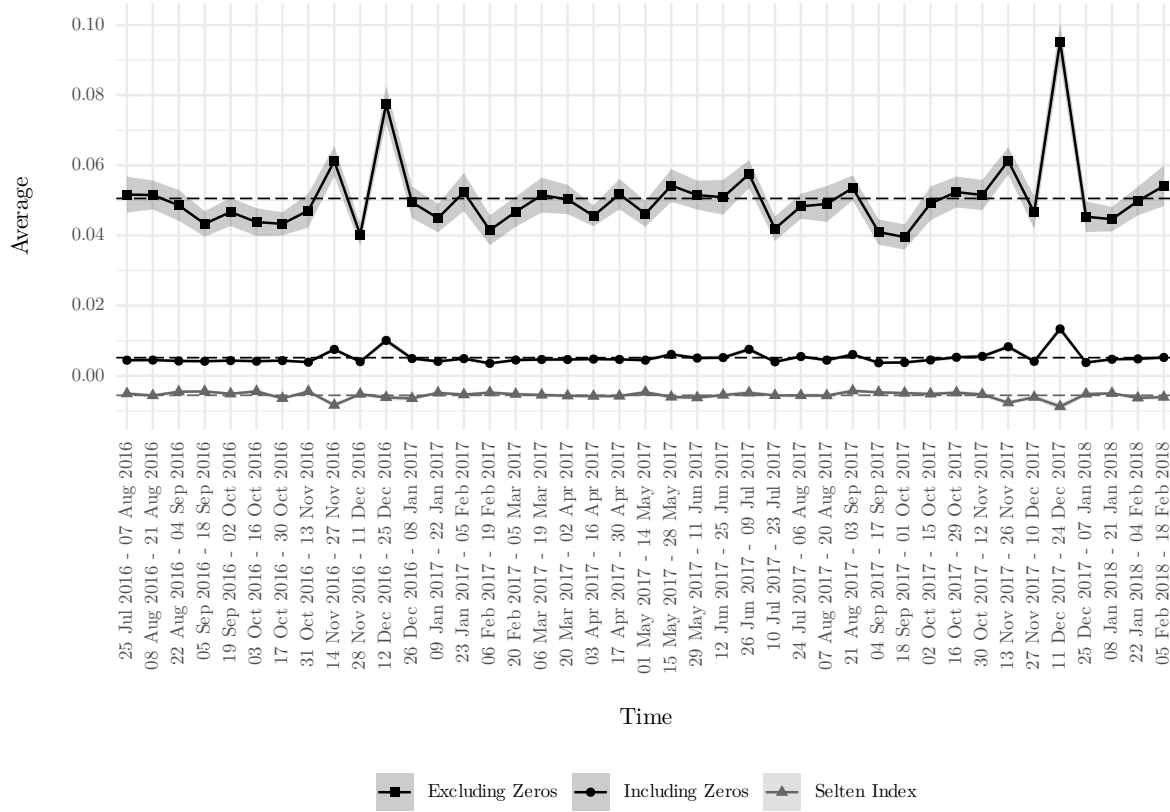


Figure 2: Cost efficiency and Selten score over time, averaged across households. Shaded areas represent ± 1 standard error around the mean.

rationality? This question can be analyzed by i) averaging over households for each time period, and doing a time series analysis, and ii) taking a dynamic panel data approach to investigate whether lags of rationality explain future periods of rationality.

For the time series analysis, for both the CEI and Selten score, we create new datasets in which we take averages over household for each time period. For the CEI, we first removed those who exhibit perfect rationality in all periods. This gives us time series of average rationality (41 periods). Using Augmented Dickey-Fuller (ADF) tests, we reject non-stationarity in levels at 10% significance (0.06 p-value) and first-differences at 5% significance. In order to determine the appropriate lagged variables in an ARIMA($p, 0, q$) model for both levels and first-differences, we plot the autocorrelation functions (ACFs) and the partial autocorrelation functions (PACFs) in Figure A1. For levels, both the ACF and PACF do not show any statistically significant correlations (as indicated by the blue confidence intervals), suggesting that the appropriate AR and MA orders are 0. This suggests use of an ARIMA(0,0,0) model which simply implies white noise i.e. each time series of rationality is best modeled as noise with no discernible pattern.¹¹ For first-differences, both ACFs exhibit a massive negative spike at the first lag (and decaying PACFs), which is strongly indicative of over-differencing, implying that the first-difference transformation is inappropriate. Overall, the time series analysis shows that both CEI and Selten scores behave as good as white noise about their means.

We estimate dynamic panel data models to investigate the effect of one (or two) lag(s) of CE on current period CE, and equivalently with the Selten score. We estimate both difference GMM (Arellano and Bond, 1991) and system GMM (Blundell and Bond, 1998).¹² For all specifications, we do two-step GMM including both individual fixed effects and time fixed effects, using robust standard errors; to avoid overfitting, we use collapsed instruments.¹³ Across the complete set of estimation results presented in Tables A9 and A10, the lagged coefficients are mostly close to zero and statistically

¹¹Instead of the ACF/PACF, another common method for determining an ARIMA model is to use information criterion. As these methods are based on different decision rules, they need not coincide with each other. We use the `auto.arima` command in R, which is based on the algorithm in Hyndman and Khandakar (2008) for choosing the optimal lags (via grid search) using the corrected AIC. In our case, this method and the ACF/PACF coincide in their suggestions for ARIMA models. This is true whether we include or exclude the display promos control variables. Using this methodology, we also conduct time series analysis at the individual level for the CEI where 89% of households have a predicted ARIMA(0,0,0), 6% with an MA(1) with the remaining households having diverse and non-standard lag structures; a similar pattern appears when analyzing the Selten score.

¹²Given the low persistence in CEI and Selten score (with AR(1) coefficients -0.0171 and -0.0137, respectively), difference GMM is a somewhat preferred specification. Before estimation, we test for unit roots with 1 lag using both the Levin-Lin-Chu (Levin, Lin, and Chu, 2002) and Maddala-Wu (Maddala and Wu, 1999) tests, with both rejecting non-stationarity at all reasonable significance levels.

¹³Roodman (2009) recommends collapsing the instrument matrix to mitigate overfitting of endogenous variables, particularly relevant in panels with a moderately large number of time periods.

insignificant. This initially suggests a clear absence of any immediate effect on rationality. However, when focusing on the system GMM including controls (demographics and display promos), statistically significant and positive first- and second-order lag effects consistently emerge for both the CEI and the Selten score. Although this implies that household behavior is somewhat driven by previous periods of rationality, the magnitude of this effect is limited. For example, holding all else equal, on average, if a household were to exhibit a maximum drop in rationality at period $t = \tau$, this implies an increase of only 5.1 percentage points for the CEI, by period $t = \tau + 1$.^{14,15}

1.5 Which types of good are responsible for irrationality?

Using the CEI, we are able to do a decomposition of irrationality by goods. Consider Example 2 in which we have 2 goods, we have

$$\begin{aligned}
 CE_1 &= \frac{\mathbf{p}^1 \cdot (\mathbf{x}^1 - \mathbf{x}^2)}{\mathbf{p}^1 \cdot \mathbf{x}^1} \\
 &= \frac{\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 - 5 \\ 5 - 2 \end{pmatrix}}{12} \\
 &= \underbrace{-\frac{3}{12}}_{\text{good 1}} + \underbrace{\frac{6}{12}}_{\text{good 2}} \\
 &= 0.25
 \end{aligned} \tag{1}$$

Recalling that $CEI_1 = 0$ means perfect rationality, observing such choices means that even though the consumer has made (relative) savings from good 1, there has been overspending on good 2, resulting in $CE_1 > 0$; in other words, good 2 is responsible for the increase in irrationality.

We calculate the relative savings for each category of goods to identify which categories are most frequently responsible for the largest increases in irrationality (i.e., the categories exhibiting the maximum overspending). Aggregating across all households

¹⁴This assumes a theoretical maximum shock of 1.0 (a 100 percentage point jump) in the Cost Efficiency Index at period $t = \tau$.

¹⁵From each table, the Sargan test p-values indicate that the instruments are valid. The p-values for the autocorrelation tests are also expected and unproblematic; the implication being that we have the expected first-order serial correlation with second-order (differenced) serial correlation not present with the indication of exogenous instruments. However, when doing Diff-GMM with an additional lag, the AR(1) p-values were large and hence omitted.

and observations, we observe 10,159 such instances with the top four being meat and seafood, dairy, non-sugary beverages, and sugary beverages. The entire set of these frequencies over time is tabulated in Table A11. Additionally, Table A12 captures the frequencies of all categories that contributed toward irrationality (i.e., instances of *any* overspending). Under this broader measure, the top four most frequent contributors are dairy, starches, sugary beverages, and fruits and vegetables, with 34,679 total instances overall.

Although there may exist intrinsic characteristics that are potentially associated with increases in consumer irrationality (e.g., highly palatable, convenient, etc.), product placement within the stores may also have an (adverse) effect on rationality (Schweiger, Ahlbom, Nordfält, Roggeveen, and Grewal, 2023; Chandon, Hutchinson, Bradlow, and Young, 2009). In our data, we have longitudinal data on 6 broad categories for in-store display (other than regular product placement), store front, store rear, end caps (front, mid-aisle, rear, side-aisle), in-aisle, secondary location display, and in-shelf.¹⁶ Table A13 tabulates the frequency with which categories over time were on non-regular display, with the categories that make the most use of them being dairy, starches, and sugary beverages. These remain the top three categories when focusing on instances where a category was responsible for the largest increase in irrationality (Table A14).

We test the association between consumer irrationality (defined as instances of overspending) and display placement using fixed effects (household, category, time) regressions.¹⁷ We do this for all instances of overspending, as well as the most severe cases. Table A16 presents the results of regressing irrationality on a dummy capturing whether a product was featured in any non-regular display location. Without category interactions, the coefficients are generally negative but not statistically significant. However, introducing interactions with product categories (with dairy omitted as the baseline) allows us to observe interesting heterogeneous effects. Focusing on the worst overspending, in combination with the positive and significant baseline coefficient on any display,

¹⁶Store front/rear displays are positioned at the entrance/back of the stores. End caps are the fixtures at the end of each aisle, further specified by their location within the store. Secondary location displays are where a product is placed outside of its natural category (incongruent placement), typically to encourage impulse buys (with natural complements), e.g., placing nuts and seeds next to beer. In-aisle refers to special displays placed within the product’s ordinary aisle (in-shelf) but separate from regular placement.

¹⁷Standard errors are two-way clustered at the category and period levels, corresponding to the level of non-regular display assignment (Abadie, Athey, Imbens, and Wooldridge, 2023). As these displays are determined by the retailer at the category-time level rather than the household level, clustering along these dimensions properly captures the shared shocks across time and category (e.g. market shocks). We omit household-level clustering because the included household fixed effects adequately absorb persistent within-household unobservable heterogeneity. Furthermore, attempting three-way clustering creates highly sparse intersection groups, which yields a non-positive semi-definite variance-covariance matrix and thus unreliable standard errors (Cameron, Gelbach, and Miller, 2011).

non-regular displays increase the probability of a good contributing to irrationality for certain categories. For example, the net effect of any display on the probability of irrationality is positive for sugary beverages ($2.97 + 2.05 > 0$) and savory snacks, with corresponding increases in the actual magnitude of irrationality relative to the baseline. Conversely, displays actually decrease both the likelihood and magnitude of irrationality relative to the baseline for categories like fruits and vegetables. These patterns remain when we consider the sample of all goods. Table A15 disaggregates these effects by specific display types, highlighting interesting patterns depending on the placement. For instance, store front placements reveal a massive reversal depending on the product type: placing goods such as dairy and fruits and vegetables at the store front actually significantly decreases both the probability and magnitude of irrationality. However, placing goods like sugary beverages and savory snacks in the same high-visibility location has the opposite effect, being associated with an increased likelihood of irrational purchasing, as well as an increased magnitude of overspending.

Although these regressions only establish correlations, the observed patterns are intuitive. Retailers naturally use prominent locations like store fronts and end caps to maximize visibility and capture shoppers' attention (Dhar, Hoch, and Kumar, 2001). These placements are highly effective at triggering impulsive decisions, especially when stocked with temptation-prone product types (Tan, Corsi, Cohen, Sharp, Lockshin, Caruso, and Bogomolova, 2018). Other marketing literature demonstrates that display effectiveness varies significantly based on both the type of display and its spatial distance from a product's primary location (Inman, Winer, and Ferraro, 2009; Han, Chandukala, and Li, 2022). While we cannot make strict causal claims, identifying these heterogeneous, category-specific relationships further highlights the empirical value of our theoretical framework by introducing a decomposition of aggregate consumer irrationality down to the level of individual goods.

1.6 What are the behavioral effects of price increases in irrational goods?

1.6.1 Welfare implications

In order to reduce the consumption of sugar, soda taxes have been widely adopted across the globe, receiving much attention in recent literature (Fletcher, Frisvold, and Tefft, 2015; Allcott, Lockwood, and Taubinsky, 2019b,a; Dubois, Griffith, and O'Connell, 2020; Shen, Wang, Yang, and An, 2023). As we are unrestricted in our welfare analysis insofar as we can calculate the (non-strict and strict) compensating variation (CV) for any form

of price change, we follow Dubois et al. (2020) and illustrate the welfare implications of a 25% tax on sugary beverages.

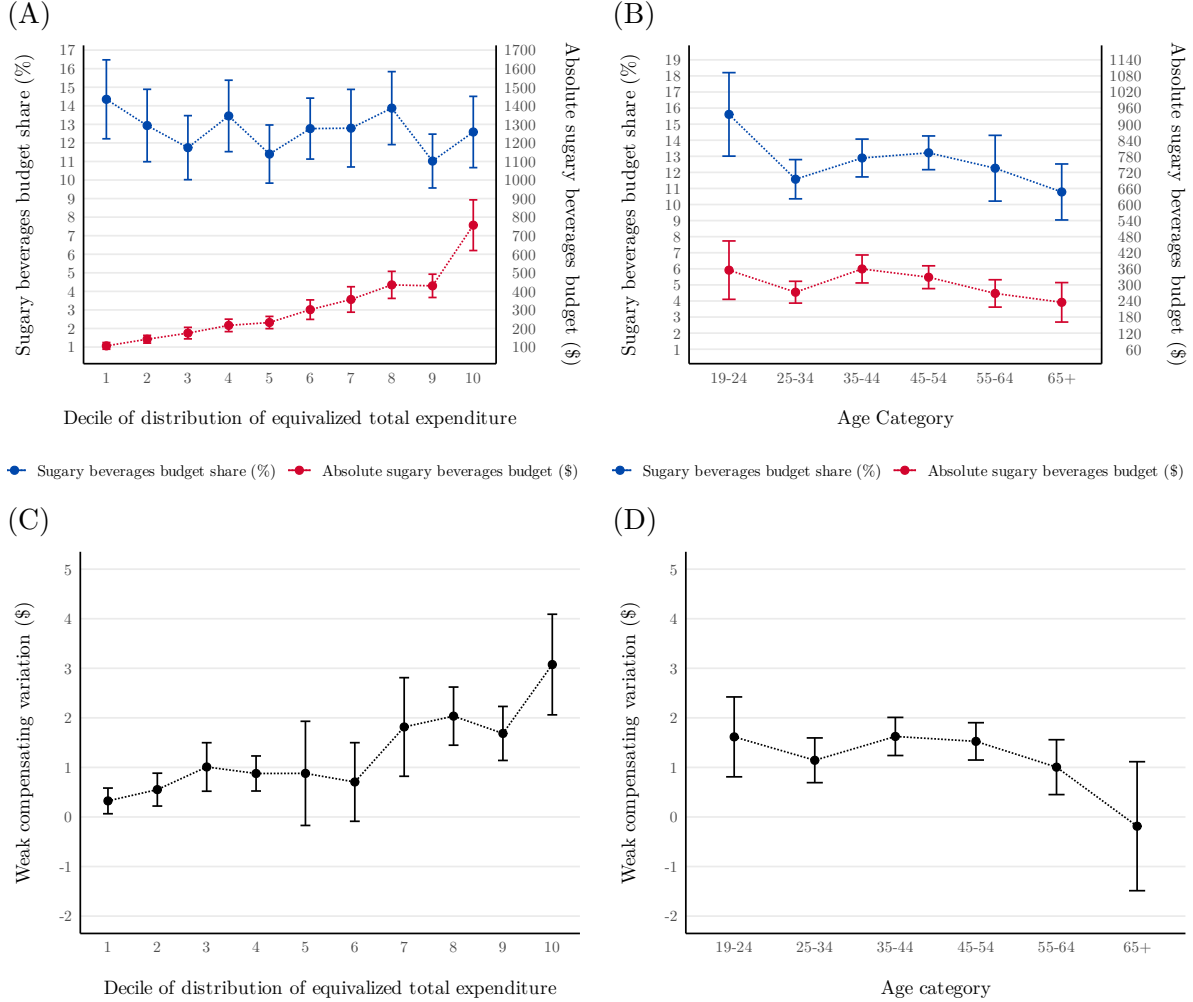


Figure 3: Weak compensating variation by age and equivalized expenditure

We analyze how compensating variation varies with age and observed equivalized expenditure. We continue the analysis assuming that observed expenditure is a reasonable proxy for income (as is standard in the literature).^{18,19} Figure 3, panel D plots average CV by age category, revealing that younger households are disproportionately

¹⁸The reported income bands are weakly positively rank-correlated with expenditure. See Figures A4 and A5 for the equivalent figures using reported income (weak and strict CV).

¹⁹As we can infer the number of adults and children within each household from the demographic data, we calculate equivalized expenditure assuming a within-household sharing rule based on the OECD equivalence scale. For each household, an equivalence value is assigned to each family member. The first adult is assigned 1.0, subsequent adults (older than 13) are assigned 0.5, and children are each assigned 0.3. Multiplying expenditure by the reciprocal of the sum of these equivalence values allows us to define equivalized expenditure for each household. See Chapters 3 and 8 of OECD (2013) for further details.

affected by the tax compared to oldest households. Panel C plots how CV changes by equivalized total expenditure, indicating that higher-expenditure households require more compensation in response to the tax increase. A common concern about excise taxes is that they are regressive, which is typically based on the observation that lower-income households tend to be relatively heavy consumers of the taxed products and bear a larger absolute monetary burden. In our setting, we observe the opposite pattern, as poorer households bear a smaller absolute monetary burden than richer households. In the absence of behavioral effects (i.e., how much households value substitute goods, and how willing consumers are to switch away from taxed goods), these relative welfare effects would follow the same pattern, given that baseline spending on sugary beverages is higher for both younger and wealthier households (Panels A and B).^{20,21} Assuming that the CV captures the total effect of the tax on welfare, Panels C and D show that even after accounting for these behavioral effects, most households are worse off, in particular, households not in the oldest cohort and those in the upper tail of the equivalized expenditure distribution.

1.6.2 Demand response

Figure A6 visualizes the estimated nonparametric lower and upper bounds of counterfactual demand for each household (represented by individual horizontal lines) under a 25% tax on sugary beverages, with observed consumption from the last period represented by a solid dot. The left column presents these bounds in quantities, whereas the right column expresses them as a percentage of the household's income.

The upper panels (above the solid line) display all households grouped by whether they satisfy or violate SARP, sorted vertically by their overall estimated bound width (i.e., the span between the upper and lower bound). Our estimates reflect a typical characteristic of nonparametric revealed preference analysis insofar as the counterfactual bounds for many households are wide. It is well established in the literature that standard methods (Varian, 1982) for demand estimation (and other related objects of interest) often yield wide and thus uninformative bounds in the absence of further restrictions (e.g., Varian, 1982; Blundell, Browning, and Crawford, 2003, 2008; Blundell, Browning, Cherchye, Crawford, De Rock, and Vermeulen, 2015). In avoiding additional

²⁰In Dubois et al. (2020), higher total expenditure is correlated with lower sugary beverage consumption, resulting in lower CV for higher-expenditure households. In our data, we observe the opposite spending pattern, meaning that the association between sugary beverage expenditure and CV in our results is consistent with their findings, robust to behavioral adjustments.

²¹To gauge magnitude, the average CV is approximately \$1.30, which is approximately 3.6% of average per-period expenditure. Figure A3 presents the equivalent panels but with CV as a percentage of expenditure.

assumptions and relying on just finite price-quantity datasets, our methodology naturally inherits this issue. That being said, current methods also theoretically require exact utility maximization vis-a-vis adherence to SARP/GARP. Through our coarse utility maximization framework, we extend the revealed preference literature by recovering demand bounds for all households (who still have well-behaved utility functions), SARP violation or not.

Despite the conservative nature of nonparametric estimation described above, we are still able to recover tight demand intervals for a number of households, as displayed in the ‘zoomed-in lower panels’ (below the solid line) of Figure A6. Specifically, we observe: (i) looking at the tightest 5% of bound widths among households with a zero lower bound, their predictive intervals are remarkably constrained relative to the majority of households; and (ii) interestingly, whenever a household’s estimated lower bound is strictly positive, the predicted demand bounds are typically drastically tighter. These observations remain true even when the last period demand is much larger than 0. This demonstrates that, for these particular households, observed historical price variation is sufficient to constrain counterfactual behavior without requiring the imposition of further identifying restrictions, even when those households violate SARP.

A Tables and Figures

	(1-)C-AEI	C-MPI	(1-)C-AVI	(1-)C-HMI
Intercept	9.035*** (0.759)	0.790*** (0.119)	14.985*** (0.934)	0.772*** (0.102)
Age: 35-54	-0.840 (0.682)	0.007 (0.097)	-0.627 (0.886)	-0.004 (0.083)
Age: 55+	0.885 (1.098)	0.244 (0.176)	1.871 (1.273)	0.252 (0.170)
Income: Middle	-0.083 (0.637)	-0.110 (0.101)	0.398 (0.824)	-0.082 (0.085)
Income: Upper	-1.139 (1.022)	-0.100 (0.181)	2.062 (1.380)	-0.024 (0.189)
Homeowner: Owner	-0.795 (0.725)	-0.108 (0.116)	-0.652 (0.965)	-0.136 (0.105)
Homeowner: Renter	0.086 (1.354)	-0.164 (0.139)	0.548 (1.560)	-0.112 (0.133)
Household Size: 3-4	0.060 (0.715)	0.047 (0.101)	-0.280 (0.940)	0.015 (0.082)
Household Size: 5+	1.060 (1.212)	0.013 (0.142)	-2.410* (1.338)	0.026 (0.132)
Marital Status: Married	-0.091 (0.677)	0.001 (0.096)	0.518 (0.904)	0.008 (0.082)
Marital Status: Single	-0.792 (0.820)	-0.047 (0.123)	-0.328 (1.204)	-0.064 (0.101)
R ²	0.013	0.009	0.015	0.012
Adj. R ²	0.000	-0.004	0.002	-0.001
Num. obs.	801	801	801	801

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A1: Multiple Regression Results (Robust SEs)

	(1-)AEI	(A)MPI	(1-)AVI	(1-)HMI
Intercept	3.493*** (0.350)	5.825*** (0.487)	6.970*** (0.401)	0.188*** (0.021)
Age: 35-54	-0.233 (0.308)	-0.480 (0.435)	-0.490 (0.378)	-0.022 (0.018)
Age: 55+	0.245 (0.460)	0.516 (0.696)	0.444 (0.518)	0.017 (0.028)
Income: Middle	-0.253 (0.287)	-0.088 (0.419)	0.154 (0.338)	-0.013 (0.017)
Income: Upper	-0.284 (0.440)	-0.697 (0.655)	0.917* (0.542)	-0.004 (0.029)
Homeowner: Owner	-0.214 (0.343)	-0.510 (0.470)	-0.345 (0.401)	-0.025 (0.019)
Homeowner: Renter	-0.163 (0.559)	-0.127 (0.814)	0.165 (0.652)	-0.036 (0.030)
Household Size: 3-4	-0.031 (0.347)	-0.002 (0.476)	-0.185 (0.402)	-0.002 (0.020)
Household Size: 5+	0.103 (0.494)	0.705 (0.808)	-0.792 (0.593)	-0.001 (0.030)
Marital Status: Married	-0.026 (0.337)	0.043 (0.448)	0.535 (0.374)	0.018 (0.018)
Marital Status: Single	-0.251 (0.384)	-0.541 (0.523)	0.020 (0.509)	-0.006 (0.021)
R ²	0.006	0.011	0.015	0.010
Adj. R ²	-0.007	-0.002	0.003	-0.002
Num. obs.	801	801	801	801

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A2: Multiple Regression Results (Robust SEs)

	SS for (1-)C-AEI	SS for C-MPI	SS for (1-)C-AVI	SS for (1-)C-HMI
Intercept	-5.810*** (0.700)	-1.180*** (0.122)	-7.471*** (0.836)	-0.894*** (0.103)
Age: 35-54	-0.806 (0.623)	0.028 (0.105)	-1.308* (0.791)	-0.030 (0.088)
Age: 55+	0.510 (1.023)	0.091 (0.190)	-0.978 (1.147)	0.038 (0.174)
Income: Middle	0.922 (0.584)	0.028 (0.108)	-0.045 (0.723)	0.025 (0.089)
Income: Upper	1.381 (0.986)	0.353* (0.186)	2.573** (1.136)	0.329* (0.187)
Homeowner: Owner	-1.015 (0.674)	-0.263** (0.126)	-1.995** (0.810)	-0.231** (0.112)
Homeowner: Renter	-0.801 (1.213)	-0.426** (0.189)	-2.090 (1.473)	-0.359** (0.155)
Household Size: 3-4	0.508 (0.661)	0.210* (0.111)	0.671 (0.834)	0.143 (0.090)
Household Size: 5+	1.495 (1.166)	0.144 (0.159)	-0.381 (1.344)	0.161 (0.147)
Marital Status: Married	0.762 (0.646)	0.108 (0.110)	0.973 (0.797)	0.106 (0.094)
Marital Status: Single	-0.265 (0.733)	0.037 (0.133)	-0.186 (0.971)	-0.020 (0.108)
R ²	0.016	0.016	0.021	0.019
Adj. R ²	0.004	0.004	0.009	0.007
Num. obs.	801	801	801	801

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A3: Multiple Regression Results with Selten Scores (SS) (Robust SEs)

	SS for (1-)AEI	SS for MPI	SS for (1-)AVI	SS for (1-)HMI
Intercept	−2.639*** (0.345)	−3.678*** (0.457)	−3.099*** (0.372)	−0.191*** (0.021)
Age: 35-54	−0.222 (0.297)	−0.457 (0.398)	−0.548 (0.354)	−0.019 (0.018)
Age: 55+	0.140 (0.450)	0.388 (0.649)	−0.407 (0.491)	−0.007 (0.028)
Income: Middle	0.187 (0.274)	0.601 (0.384)	−0.022 (0.310)	0.009 (0.017)
Income: Upper	0.800* (0.432)	0.942 (0.632)	1.253** (0.486)	0.070** (0.029)
Homeowner: Owner	−0.296 (0.332)	−0.578 (0.436)	−0.808** (0.348)	−0.040** (0.020)
Homeowner: Renter	−0.514 (0.548)	−0.569 (0.748)	−0.349 (0.657)	−0.059* (0.033)
Household Size: 3-4	0.161 (0.336)	0.271 (0.440)	0.249 (0.379)	0.022 (0.020)
Household Size: 5+	0.296 (0.493)	0.945 (0.768)	−0.249 (0.584)	0.015 (0.031)
Marital Status: Married	0.344 (0.331)	0.585 (0.425)	0.567* (0.344)	0.037* (0.019)
Marital Status: Single	−0.019 (0.364)	−0.181 (0.466)	−0.029 (0.429)	0.005 (0.021)
R ²	0.010	0.016	0.022	0.022
Adj. R ²	−0.002	0.004	0.010	0.010
Num. obs.	801	801	801	801

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A4: Multiple Regression Results with Selten Scores (Robust SEs)

	OLS	RE	OLS	RE	OLS	RE	OLS	RE
Intercept	0.4481*** (0.0205)	0.4481*** (0.0205)	0.7014*** (0.0982)	0.7014*** (0.0982)	0.4393*** (0.0218)	0.4481*** (0.0217)	0.7429*** (0.1038)	0.7206*** (0.1006)
Time	0.0033*** (0.0007)	0.0033*** (0.0007)	0.0033*** (0.0013)	0.0033*** (0.0013)	0.0032*** (0.0007)	0.0032*** (0.0007)	0.0033*** (0.0013)	0.0034*** (0.0013)
LogL	-244155	-241004	-81321	-79882	-244146	-241000	-81307	-79878
LM-test (OLS)		0		0		0		0
Num. Obs.	101434	101434	32841	32841	101434	101434	32841	32841
Num. Groups		2474		801		2474		801
Var(Intercept)		0.7101		1.0446		0.7099		1.0412
Var(residual)	7.2153	6.5051	8.2886	7.2410	7.2148	6.5046	8.2839	7.2393
Demographics	No	No	Yes	Yes	No	No	Yes	Yes
Display Promos	No	No	No	No	Yes	Yes	Yes	Yes
AIC	488316	482015	162667	159793	488317	482026	162658	159801
BIC	488344	482053	162777	159910	488431	482150	162843	159994

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A5: Regression table for CEI and a time trend; RE is the random effects model.

	OLS	RE	OLS	RE	OLS	RE	OLS	RE
Intercept	-0.5215*** (0.0229)	-0.5215*** (0.0229)	-0.8030*** (0.1008)	-0.8030*** (0.1008)	-0.4077*** (0.0244)	-0.4170*** (0.0240)	-0.7373*** (0.1060)	-0.6839*** (0.1038)
Time	-0.0017** (0.0007)	-0.0017** (0.0007)	-0.0043*** (0.0015)	-0.0043*** (0.0015)	-0.0018** (0.0007)	-0.0018** (0.0007)	-0.0043*** (0.0014)	-0.0043*** (0.0014)
LogL	-246924	-242838	-82680	-81226	-246575	-242601	-82656	-81162
LM-test (OLS)		0		0		0		0
Num. Obs.	101434	101434	32841	32841	101434	101434	32841	32841
Num. Groups		2474		801		2474		801
Var(Intercept)		0.9042		1.1437		0.8806		1.1756
Var(residual)	7.6203	6.7159	9.0040	7.8570	7.5687	6.6878	8.9932	7.8211
Demographics	No	No	Yes	Yes	No	No	Yes	Yes
Display Promos	No	No	No	No	Yes	Yes	Yes	Yes
AIC	493855	485683	165386	162480	493175	485228	165356	162370
BIC	493883	485721	165495	162598	493289	485352	165541	162563

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A6: Regression table for SS and a time trend; RE is the random effects model.

	OLS	RE	OLS	RE	OLS	RE	OLS	RE
Dummy variables								
25/07/16 - 07/08/16	-0.0716 (0.0528)	-0.0716 (0.0528)	0.3882** (0.1907)	0.3882** (0.1907)	-0.0798 (0.0539)	-0.0698 (0.0534)	0.4317** (0.1947)	0.4076** (0.1924)
08/08/16 - 21/08/16	-0.0684 (0.0462)	-0.0684 (0.0462)	0.4439** (0.2079)	0.4439** (0.2078)	-0.0768* (0.0461)	-0.0670 (0.0462)	0.4774** (0.2096)	0.4598** (0.2083)
22/08/16 - 04/09/16	-0.0941** (0.0471)	-0.0941** (0.0471)	0.3797* (0.2236)	0.3797* (0.2235)	-0.1041** (0.0483)	-0.0932* (0.0479)	0.4142* (0.2269)	0.3963* (0.2253)
05/09/16 - 18/09/16	-0.1015** (0.0431)	-0.1015** (0.0431)	0.3721* (0.1964)	0.3721* (0.1963)	-0.1102** (0.0435)	-0.1011** (0.0432)	0.4003** (0.1985)	0.3842* (0.1970)
19/09/16 - 02/10/16	-0.0841* (0.0461)	-0.0841* (0.0461)	0.4380** (0.1920)	0.4380** (0.1920)	-0.0918** (0.0462)	-0.0827* (0.0463)	0.4805** (0.1954)	0.4581** (0.1937)
03/10/16 - 16/10/16	-0.1002** (0.0455)	-0.1002** (0.0455)	0.5377** (0.2178)	0.5377** (0.2177)	-0.1092** (0.0458)	-0.0999** (0.0461)	0.5727*** (0.2212)	0.5527** (0.2194)
17/10/16 - 30/10/16	-0.0838** (0.0421)	-0.0838** (0.0421)	0.4315* (0.2211)	0.4315* (0.2210)	-0.0913** (0.0427)	-0.0835* (0.0428)	0.4632** (0.2240)	0.4450** (0.2226)
31/10/16 - 13/11/16	-0.1279*** (0.0475)	-0.1279*** (0.0475)	0.3147* (0.1899)	0.3147* (0.1898)	-0.1349*** (0.0481)	-0.1272*** (0.0477)	0.3522* (0.1927)	0.3321* (0.1911)
14/11/16 - 27/11/16	0.2358*** (0.0661)	0.2358*** (0.0661)	0.7970*** (0.2095)	0.7970*** (0.2095)	0.2282*** (0.0668)	0.2365*** (0.0670)	0.8333*** (0.2136)	0.8137*** (0.2124)
28/11/16 - 11/12/16	-0.1133*** (0.0397)	-0.1133*** (0.0397)	0.3517** (0.1778)	0.3517** (0.1777)	-0.1219*** (0.0401)	-0.1133*** (0.0398)	0.3845** (0.1803)	0.3664** (0.1789)
12/12/16 - 25/12/16	0.4917*** (0.0855)	0.4917*** (0.0855)	1.0121*** (0.2300)	1.0121*** (0.2300)	0.4832*** (0.0864)	0.4913*** (0.0859)	1.0465*** (0.2337)	1.0269*** (0.2319)
26/12/16 - 08/01/17	-0.0245 (0.0537)	-0.0245 (0.0537)	0.6336*** (0.2435)	0.6336*** (0.2434)	-0.0356 (0.0544)	-0.0249 (0.0549)	0.6670*** (0.2468)	0.6494*** (0.2454)
09/01/17 - 22/01/17	-0.1058** (0.0451)	-0.1058** (0.0451)	0.4457** (0.1960)	0.4457** (0.1959)	-0.1152** (0.0449)	-0.1057** (0.0449)	0.4841** (0.1976)	0.4628** (0.1962)
23/01/17 - 05/02/17	-0.0274 (0.0591)	-0.0274 (0.0591)	0.4419** (0.2046)	0.4419** (0.2045)	-0.0392 (0.0601)	-0.0278 (0.0602)	0.4790** (0.2089)	0.4578** (0.2067)
06/02/17 - 19/02/17	-0.1623*** (0.0432)	-0.1623*** (0.0432)	0.4304* (0.2402)	0.4304* (0.2401)	-0.1718*** (0.0432)	-0.1627*** (0.0436)	0.4606* (0.2422)	0.4439* (0.2411)
20/02/17 - 05/03/17	-0.0665 (0.0495)	-0.0665 (0.0495)	0.3845* (0.2033)	0.3845* (0.2032)	-0.0738 (0.0501)	-0.0650 (0.0498)	0.4250** (0.2062)	0.4041** (0.2047)
06/03/17 - 19/03/17	-0.0522 (0.0538)	-0.0522 (0.0538)	0.5091** (0.2476)	0.5091** (0.2475)	-0.0583 (0.0540)	-0.0506 (0.0541)	0.5533** (0.2501)	0.5294** (0.2487)
20/03/17 - 02/04/17	-0.0530 (0.0484)	-0.0530 (0.0484)	0.4704** (0.1908)	0.4704** (0.1907)	-0.0604 (0.0492)	-0.0514 (0.0490)	0.5147*** (0.1943)	0.4917** (0.1925)
03/04/17 - 16/04/17	-0.0410 (0.0419)	-0.0410 (0.0419)	0.4017** (0.2049)	0.4017** (0.2048)	-0.0494 (0.0426)	-0.0399 (0.0430)	0.4465** (0.2089)	0.4228** (0.2077)
17/04/17 - 30/04/17	-0.0497 (0.0491)	-0.0497 (0.0491)	0.4595** (0.1909)	0.4595** (0.1909)	-0.0565 (0.0493)	-0.0481 (0.0495)	0.5035*** (0.1940)	0.4803** (0.1927)
01/05/17 - 14/05/17	-0.0704 (0.0451)	-0.0704 (0.0450)	0.4045* (0.2157)	0.4045* (0.2156)	-0.0754* (0.0454)	-0.0680 (0.0454)	0.4474** (0.2183)	0.4246* (0.2171)
15/05/17 - 28/05/17	0.0930 (0.0625)	0.0930 (0.0625)	0.3988** (0.2025)	0.3988** (0.2024)	0.0858 (0.0636)	0.0947 (0.0633)	0.4422** (0.2066)	0.4218** (0.2054)
29/05/17 - 11/06/17	-0.0105 (0.0508)	-0.0105 (0.0508)	0.4535** (0.2027)	0.4535** (0.2027)	-0.0176 (0.0510)	-0.0096 (0.0511)	0.4894** (0.2046)	0.4695** (0.2037)
12/06/17 - 25/06/17	0.0021 (0.0591)	0.0021 (0.0591)	0.4284** (0.1832)	0.4284** (0.1831)	-0.0056 (0.0599)	0.0017 (0.0597)	0.4638** (0.1860)	0.4441** (0.1849)
26/06/17 - 09/07/17	0.2366*** (0.0652)	0.2366*** (0.0651)	0.6908*** (0.2042)	0.6908*** (0.2041)	0.2265*** (0.0659)	0.2369*** (0.0655)	0.7326*** (0.2086)	0.7097*** (0.2059)
10/07/17 - 23/07/17	-0.1180*** (0.0414)	-0.1180*** (0.0414)	0.4114** (0.2083)	0.4114** (0.2082)	-0.1264*** (0.0420)	-0.1177*** (0.0422)	0.4449** (0.2120)	0.4263** (0.2104)
24/07/17 - 06/08/17	0.0324 (0.0509)	0.0324 (0.0509)	0.5227** (0.2054)	0.5227** (0.2054)	0.0224 (0.0520)	0.0329 (0.0514)	0.5679*** (0.2091)	0.5438*** (0.2066)
07/08/17 - 20/08/17	-0.0709 (0.0544)	-0.0709 (0.0544)	0.2965 (0.1936)	0.2965 (0.1936)	-0.0781 (0.0555)	-0.0691 (0.0549)	0.3423* (0.1965)	0.3184 (0.1945)
21/08/17 - 03/09/17	0.0898* (0.0530)	0.0898* (0.0530)	0.6302*** (0.2057)	0.6302*** (0.2057)	0.0820 (0.0539)	0.0917* (0.0535)	0.6608*** (0.2098)	0.6404*** (0.2073)
04/09/17 - 17/09/17	-0.1453*** (0.0403)	-0.1453*** (0.0403)	0.4149** (0.1937)	0.4149** (0.1936)	-0.1527*** (0.0411)	-0.1442*** (0.0409)	0.4492** (0.1972)	0.4304** (0.1954)
18/09/17 - 01/10/17	-0.1360*** (0.0419)	-0.1360*** (0.0418)	0.4226** (0.1973)	0.4226** (0.1972)	-0.1431*** (0.0427)	-0.1352*** (0.0426)	0.4580** (0.2004)	0.4397** (0.1986)
02/10/17 - 15/10/17	-0.0655 (0.0524)	-0.0655 (0.0524)	0.3249 (0.2099)	0.3249 (0.2098)	-0.0727 (0.0529)	-0.0649 (0.0529)	0.3580* (0.2127)	0.3405 (0.2113)
16/10/17 - 29/10/17	0.0109 (0.0546)	0.0109 (0.0546)	0.6650*** (0.2250)	0.6650*** (0.2250)	0.0043 (0.0549)	0.0121 (0.0550)	0.6987*** (0.2280)	0.6823*** (0.2264)
30/10/17 - 12/11/17	0.0377 (0.0564)	0.0377 (0.0564)	0.5282** (0.2106)	0.5282** (0.2105)	0.0305 (0.0573)	0.0388 (0.0571)	0.5580*** (0.2140)	0.5425** (0.2121)
13/11/17 - 26/11/17	0.3108*** (0.0677)	0.3108*** (0.0677)	0.9090*** (0.2580)	0.9090*** (0.2579)	0.3013*** (0.0676)	0.3105*** (0.0679)	0.9423*** (0.2593)	0.9249*** (0.2582)
27/11/17 - 10/12/17	-0.1043** (0.0491)	-0.1043** (0.0491)	0.4645** (0.1975)	0.4645** (0.1975)	-0.1125** (0.0496)	-0.1044** (0.0498)	0.4899** (0.2009)	0.4762** (0.1997)
11/12/17 - 24/12/17	0.8198*** (0.1012)	0.8198*** (0.1012)	1.5662*** (0.3147)	1.5662*** (0.3146)	0.8094*** (0.1020)	0.8192*** (0.1017)	1.6029*** (0.3177)	1.5827*** (0.3159)
25/12/17 - 07/01/18	-0.1387*** (0.0442)	-0.1387*** (0.0442)	0.4071* (0.2184)	0.4071* (0.2184)	-0.1477*** (0.0447)	-0.1388*** (0.0448)	0.4356** (0.2213)	0.4214* (0.2201)
08/01/18 - 21/01/18	-0.0455 (0.0459)	-0.0455 (0.0459)	0.3541* (0.1912)	0.3541* (0.1912)	-0.0547 (0.0466)	-0.0448 (0.0464)	0.3923** (0.1937)	0.3733* (0.1926)
22/01/18 - 04/02/18	-0.0351 (0.0492)	-0.0351 (0.0492)	0.4261** (0.2061)	0.4261** (0.2060)	-0.0469 (0.0498)	-0.0348 (0.0495)	0.4681** (0.2087)	0.4467** (0.2071)
05/02/18 - 18/02/18	0.0066 (0.0647)	0.0066 (0.0646)	0.4405** (0.2177)	0.4405** (0.2177)	-0.0023 (0.0647)	0.0077 (0.0643)	0.4740** (0.2185)	0.4567** (0.2173)
LogL	-243929	-240752	-81224	-79772	-243921	-240750	-81210	-79767
LM-test (OLS)		0		0		0		0
Num. Obs.	101434	101434	32841	32841	101434	101434	32841	32841
Num. Groups		2474		801		2474		801
Var(Interecept)		0.7109		1.0458		0.7107		1.0423
Var(residual)	7.1860	6.4721	8.2497	7.1911	7.1855	6.4718	8.2451	7.1897
Demographics	No	No	Yes	Yes	No	No	Yes	Yes
Display Promos	No	No	No	No	Yes	Yes	Yes	Yes
AIC	487941	481591	162552	159649	487944	481603	162543	159659
BIC	488341	482000	162989	160094	488430	482099	163055	160179

***p < 0.01; **p < 0.05; *p < 0.1

Table A7: Regression table for CEI, time dummies, demographics, and display indicators; RE is the random effects model

	OLS	RE	OLS	RE	OLS	RE	OLS	RE
Dummy variables								
25/07/16 - 07/08/16	0.0474 (0.0529)	0.0474 (0.0529)	-0.2388 (0.1935)	-0.2388 (0.1934)	0.1786*** (0.0538)	0.1677*** (0.0533)	-0.1757 (0.1970)	-0.1216 (0.1950)
08/08/16 - 21/08/16	-0.0066 (0.0482)	-0.0066 (0.0482)	-0.2010 (0.2121)	-0.2010 (0.2121)	0.1066** (0.0484)	0.0979** (0.0483)	-0.1487 (0.2138)	-0.1029 (0.2129)
22/08/16 - 04/09/16	0.1014** (0.0497)	0.1014** (0.0497)	-0.1141 (0.2282)	-0.1141 (0.2282)	0.2216*** (0.0511)	0.2124*** (0.0506)	-0.0602 (0.2312)	-0.0102 (0.2300)
05/09/16 - 18/09/16	0.1106** (0.0444)	0.1106** (0.0444)	-0.0953 (0.1982)	-0.0953 (0.1981)	0.2129*** (0.0447)	0.2050*** (0.0445)	-0.0517 (0.2002)	-0.0108 (0.1988)
19/09/16 - 02/10/16	0.0435 (0.0492)	0.0435 (0.0492)	-0.2506 (0.1938)	-0.2506 (0.1937)	0.1619*** (0.0494)	0.1520*** (0.0493)	-0.1896 (0.1968)	-0.1433 (0.1952)
03/10/16 - 16/10/16	0.1120** (0.0453)	0.1120** (0.0453)	0.0261 (0.2166)	0.0261 (0.2165)	0.2214*** (0.0461)	0.2132*** (0.0461)	0.0785 (0.2198)	0.1266 (0.2182)
17/10/16 - 30/10/16	-0.0873* (0.0456)	-0.0873* (0.0456)	-0.3241 (0.2215)	-0.3241 (0.2215)	0.0098 (0.0462)	0.0020 (0.0461)	-0.2782 (0.2241)	-0.2386 (0.2225)
31/10/16 - 13/11/16	0.1093** (0.0483)	0.1093** (0.0483)	-0.1246 (0.1937)	-0.1246 (0.1936)	0.2108*** (0.0490)	0.2021*** (0.0483)	-0.0707 (0.1963)	-0.0286 (0.1949)
14/11/16 - 27/11/16	-0.2809*** (0.0690)	-0.2809*** (0.0690)	-0.6160*** (0.2126)	-0.6160*** (0.2125)	-0.1745*** (0.0694)	-0.1836*** (0.0694)	-0.5619*** (0.2163)	-0.5181** (0.2148)
28/11/16 - 11/12/16	0.0329 (0.0416)	0.0329 (0.0416)	-0.2248 (0.1887)	-0.2248 (0.1887)	0.1348*** (0.0421)	0.1266*** (0.0417)	-0.1744 (0.1909)	-0.1312 (0.1897)
12/12/16 - 25/12/16	-0.0611 (0.0851)	-0.0611 (0.0851)	-0.5921** (0.2322)	-0.5921** (0.2322)	0.0449 (0.0858)	0.0355 (0.0853)	-0.5398** (0.2354)	-0.4943** (0.2334)
26/12/16 - 08/01/17	-0.0887 (0.0567)	-0.0887 (0.0567)	-0.2748 (0.2423)	-0.2748 (0.2423)	0.0245 (0.0573)	0.0164 (0.0575)	-0.2217 (0.2454)	-0.1737 (0.2438)
09/01/17 - 22/01/17	0.0705 (0.0468)	0.0705 (0.0468)	-0.1059 (0.1980)	-0.1059 (0.1979)	0.1875*** (0.0467)	0.1785*** (0.0464)	-0.0492 (0.1994)	0.0012 (0.1981)
23/01/17 - 05/02/17	0.0179 (0.0589)	0.0179 (0.0589)	-0.3045 (0.2049)	-0.3045 (0.2049)	0.1482** (0.0601)	0.1378** (0.0598)	-0.2432 (0.2093)	-0.1839 (0.2072)
06/02/17 - 19/02/17	0.0769* (0.0465)	0.0769* (0.0465)	-0.0587 (0.2409)	-0.0587 (0.2408)	0.1780*** (0.0469)	0.1707*** (0.0470)	-0.0108 (0.2431)	0.0361 (0.2419)
20/02/17 - 05/03/17	0.0303 (0.0513)	0.0303 (0.0513)	-0.3483* (0.2057)	-0.3483* (0.2057)	0.1477*** (0.0523)	0.1378*** (0.0518)	-0.2882 (0.2087)	-0.2397 (0.2073)
06/03/17 - 19/03/17	0.0088 (0.0552)	0.0088 (0.0552)	-0.2077 (0.2497)	-0.2077 (0.2496)	0.1207** (0.0553)	0.1107** (0.0552)	-0.1468 (0.2523)	-0.1011 (0.2507)
20/03/17 - 02/04/17	-0.0137 (0.0500)	-0.0137 (0.0500)	-0.3013 (0.1918)	-0.3013 (0.1917)	0.1055** (0.0506)	0.0956* (0.0503)	-0.2374 (0.1949)	-0.1887 (0.1933)
03/04/17 - 16/04/17	-0.0246 (0.0459)	-0.0246 (0.0459)	-0.3481* (0.2085)	-0.3481* (0.2084)	0.0971** (0.0463)	0.0867* (0.0466)	-0.2821 (0.2122)	-0.2311 (0.2108)
17/04/17 - 30/04/17	-0.0213 (0.0521)	-0.0213 (0.0521)	-0.3043 (0.1948)	-0.3043 (0.1947)	0.0954* (0.0525)	0.0854 (0.0523)	-0.2422 (0.1978)	-0.1928 (0.1963)
01/05/17 - 14/05/17	0.0885* (0.0471)	0.0885* (0.0471)	-0.1706 (0.2164)	-0.1706 (0.2163)	0.2048*** (0.0474)	0.1942*** (0.0473)	-0.1119 (0.2188)	-0.0656 (0.2176)
15/05/17 - 28/05/17	-0.0482 (0.0641)	-0.0482 (0.0641)	-0.6081*** (0.2028)	-0.6081*** (0.2027)	0.0665 (0.0653)	0.0570 (0.0649)	-0.5471*** (0.2067)	-0.4974** (0.2057)
29/05/17 - 11/06/17	-0.0670 (0.0523)	-0.0670 (0.0523)	-0.5270** (0.2058)	-0.5270** (0.2058)	0.0394 (0.0526)	0.0305 (0.0524)	-0.4746** (0.2079)	-0.4294** (0.2070)
12/06/17 - 25/06/17	0.0069 (0.0600)	0.0069 (0.0600)	-0.3733** (0.1869)	-0.3733** (0.1869)	0.1084* (0.0608)	0.1001* (0.0604)	-0.3239* (0.1895)	-0.2824 (0.1885)
26/06/17 - 09/07/17	0.0693 (0.0642)	0.0693 (0.0642)	-0.3788* (0.2047)	-0.3788* (0.2047)	0.1956*** (0.0651)	0.1855*** (0.0645)	-0.3144 (0.2090)	-0.2551 (0.2065)
10/07/17 - 23/07/17	-0.0032 (0.0441)	-0.0032 (0.0441)	-0.2457 (0.2099)	-0.2457 (0.2098)	0.1020** (0.0449)	0.0939** (0.0449)	-0.1942 (0.2133)	-0.1436 (0.2117)
24/07/17 - 06/08/17	-0.0051 (0.0532)	-0.0051 (0.0532)	-0.3843* (0.2103)	-0.3843* (0.2102)	0.1274** (0.0543)	0.1163** (0.0535)	-0.3144 (0.2138)	-0.2554 (0.2113)
07/08/17 - 20/08/17	-0.0088 (0.0565)	-0.0088 (0.0565)	-0.3829* (0.1996)	-0.3829* (0.1995)	0.1125* (0.0576)	0.1026* (0.0568)	-0.3156 (0.2019)	-0.2597 (0.2000)
21/08/17 - 03/09/17	0.1278** (0.0525)	0.1278** (0.0525)	-0.1957 (0.2051)	-0.1957 (0.2051)	0.2458*** (0.0529)	0.2364*** (0.0525)	-0.1329 (0.2084)	-0.0753 (0.2058)
04/09/17 - 17/09/17	0.0846** (0.0418)	0.0846** (0.0418)	-0.1484 (0.1933)	-0.1484 (0.1932)	0.1862*** (0.0425)	0.1782*** (0.0421)	-0.0981 (0.1966)	-0.0511 (0.1950)
18/09/17 - 01/10/17	0.0636 (0.0453)	0.0636 (0.0453)	-0.2414 (0.1972)	-0.2414 (0.1972)	0.1676*** (0.0459)	0.1587*** (0.0456)	-0.1877 (0.2002)	-0.1441 (0.1982)
02/10/17 - 15/10/17	0.0385 (0.0553)	0.0385 (0.0553)	-0.3354 (0.2122)	-0.3354 (0.2121)	0.1361** (0.0561)	0.1281** (0.0559)	-0.2865 (0.2147)	-0.2434 (0.2135)
16/10/17 - 29/10/17	0.0747 (0.0543)	0.0747 (0.0543)	-0.1481 (0.2296)	-0.1481 (0.2295)	0.1699*** (0.0546)	0.1622*** (0.0544)	-0.0989 (0.2322)	-0.0572 (0.2307)
30/10/17 - 12/11/17	0.0235 (0.0575)	0.0235 (0.0575)	-0.3139 (0.2136)	-0.3139 (0.2136)	0.1202** (0.0580)	0.1132* (0.0578)	-0.2706 (0.2168)	-0.2301 (0.2151)
13/11/17 - 26/11/17	-0.2117*** (0.0694)	-0.2117*** (0.0694)	-0.5164** (0.2610)	-0.5164** (0.2609)	-0.0988 (0.0691)	-0.1079 (0.0692)	-0.4617* (0.2619)	-0.4129 (0.2609)
27/11/17 - 10/12/17	-0.0552 (0.0527)	-0.0552 (0.0527)	-0.2986 (0.2035)	-0.2986 (0.2034)	0.0348 (0.0533)	0.0282 (0.0533)	-0.2576 (0.2068)	-0.2149 (0.2054)
11/12/17 - 24/12/17	-0.3208*** (0.1021)	-0.3208*** (0.1021)	-0.7665** (0.3114)	-0.7665** (0.3113)	-0.2090*** (0.1024)	-0.2178** (0.1020)	-0.7086** (0.3138)	-0.6537** (0.3116)
25/12/17 - 07/01/18	0.0369 (0.0453)	0.0369 (0.0453)	-0.1650 (0.2144)	-0.1650 (0.2143)	0.1313*** (0.0458)	0.1246*** (0.0457)	-0.1166 (0.2169)	-0.0709 (0.2155)
08/01/18 - 21/01/18	0.0554 (0.0467)	0.0554 (0.0467)	-0.2891 (0.1940)	-0.2891 (0.1940)	0.1712*** (0.0472)	0.1623*** (0.0470)	-0.2301 (0.1962)	-0.1840 (0.1951)
22/01/18 - 04/02/18	-0.0738 (0.0520)	-0.0738 (0.0520)	-0.3721* (0.2118)	-0.3721* (0.2117)	0.0599 (0.0523)	0.0496 (0.0520)	-0.3062 (0.2141)	-0.2464 (0.2125)
05/02/18 - 18/02/18	-0.0532 (0.0665)	-0.0532 (0.0665)	-0.4303* (0.2201)	-0.4303* (0.2201)	0.0578 (0.0668)	0.0499 (0.0662)	-0.3777* (0.2209)	-0.3275 (0.2198)
LogL	-246865	-242770	-82636	-81176	-246515	-242533	-82613	-81113
LM-test (OLS)		0		0		0		0
Num. Obs.	101434	101434	32841	32841	101434	101434	32841	32841
Num. Groups		2474		801		2474		801
Var(Intercept)		0.9044		1.1443		0.8808		1.1760
Var(residual)	7.6143	6.7068	8.9907	7.8325	7.5626	6.6786	8.9804	7.7973
Demographics	No	No	Yes	Yes	No	No	Yes	Yes
Display Promos	No	No	No	No	Yes	Yes	Yes	Yes
AIC	493814	485626	165377	162458	493133	485171	165348	162350
BIC	494214	486036	165814	162903	493618	485666	165861	162871

***p < 0.01; **p < 0.05; *p < 0.1

Table A8: Regression table for SS, time dummies, demographics, and display indicators; RE is the random effects model

	Diff-GMM	Diff-GMM	Diff-GMM	Diff-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM
CEI_{t-1}	-0.006	-0.006	-0.003	0.006	0.002	0.002	0.005	0.006	0.024	0.010	0.051**	0.048*
	(0.014)	(0.014)	(0.014)	(0.013)	(0.014)	(0.014)	(0.014)	(0.015)	(0.029)	(0.024)	(0.024)	(0.027)
CEI_{t-2}							0.013	0.015			0.054***	0.050**
							(0.014)	(0.015)			(0.021)	(0.023)
Demographics	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Display Promos	No	No	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
Instrument lags	2-3	2-4	2-3	2-4	2-3	2-4	3-4	3-5	2-3	2-4	3-4	3-5
Sargan p-value	0.238	0.488	0.674	0.712	0.131	0.253	0.438	0.332	0.767	0.508	0.586	0.366
AR(1) p-value	0	0	0	0	0	0	0	0	0	0	0	0
AR(2) p-value	0.384	0.383	0.342	0.206	0.454	0.457	0.757	0.818	0.115	0.064	0.523	0.498
Wald (coef) p-value	0.656	0.656	0.835	0.756	0.877	0.869	0.654	0.595	0.810	0.558	0.436	0.638
Wald (time) p-value	0	0	0	0	0	0	0	0	0	0	0	0
N	1746	1746	1746	1746	1746	1746	1746	1746	721	721	721	721

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A9: Combined Dynamic Panel Estimation Results for CEI

	Diff-GMM	Diff-GMM	Diff-GMM	Diff-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM	Sys-GMM
SS_{t-1}	-0.007	-0.007	-0.002	-0.001	0.002	0.002	0.004	0.005	0.025	0.019	0.046**	0.041*
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.025)	(0.023)	(0.023)	(0.023)
SS_{t-2}							0.014	0.015			0.045**	0.045**
							(0.013)	(0.013)			(0.019)	(0.019)
Demographics	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Display Promos	No	No	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
Instrument lags	2-3	2-4	2-3	2-4	2-3	2-4	3-4	3-5	2-3	2-4	3-4	3-5
Sargan p-value	0.173	0.385	0.375	0.764	0.070	0.148	0.382	0.313	0.806	0.640	0.810	0.743
AR(1) p-value	0	0	0	0	0	0	0	0	0	0	0	0
AR(2) p-value	0.297	0.302	0.254	0.229	0.386	0.395	0.731	0.775	0.164	0.128	0.503	0.556
Wald (coef) p-value	0.605	0.591	0.715	0.652	0.884	0.858	0.555	0.516	0.200	0.272	0.198	0.080
Wald (time) p-value	0	0	0	0	0	0	0	0	0	0	0	0
N	2337	2337	2337	2337	2337	2337	2337	2337	801	801	801	801

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Table A10: Combined Dynamic Panel Estimation Results for SS

	Dairy	Fruits and vegetables	Meat and Seafood	Non-sugary beverages	Prepared foods	Savory snacks	Starches	Sugary beverages	Sweet snacks	Row Total
25 Jul 2016 - 07 Aug 2016	27	15	69	19	22	9	13	15	7	196
08 Aug 2016 - 21 Aug 2016	22	38	45	19	34	13	12	15	12	210
22 Aug 2016 - 04 Sep 2016	29	27	66	17	20	15	16	16	9	215
05 Sep 2016 - 18 Sep 2016	49	19	64	21	17	15	15	5	11	216
19 Sep 2016 - 02 Oct 2016	15	35	38	13	9	14	11	20	44	199
03 Oct 2016 - 16 Oct 2016	19	39	57	21	17	14	14	10	11	202
17 Oct 2016 - 30 Oct 2016	39	35	85	16	11	14	15	15	23	253
31 Oct 2016 - 13 Nov 2016	26	26	41	20	8	2	13	19	18	173
14 Nov 2016 - 27 Nov 2016	88	55	46	33	8	34	49	65	32	410
28 Nov 2016 - 11 Dec 2016	32	16	61	6	9	9	30	33	39	235
12 Dec 2016 - 25 Dec 2016	81	61	33	56	8	36	34	88	46	443
26 Dec 2016 - 08 Jan 2017	33	16	49	35	6	26	31	26	27	249
09 Jan 2017 - 22 Jan 2017	25	12	45	42	11	16	14	22	7	194
23 Jan 2017 - 05 Feb 2017	14	31	39	28	20	8	14	38	10	202
06 Feb 2017 - 19 Feb 2017	21	24	44	13	15	16	20	22	11	186
20 Feb 2017 - 05 Mar 2017	17	22	71	48	11	6	6	22	17	220
06 Mar 2017 - 19 Mar 2017	13	31	75	17	11	2	6	29	9	193
20 Mar 2017 - 02 Apr 2017	22	30	43	14	16	11	10	20	46	212
03 Apr 2017 - 16 Apr 2017	39	32	17	36	13	27	29	50	22	265
17 Apr 2017 - 30 Apr 2017	24	33	63	11	35	21	17	14	12	230
01 May 2017 - 14 May 2017	31	31	63	16	7	15	14	6	11	194
15 May 2017 - 28 May 2017	44	39	58	21	29	11	35	41	7	285
29 May 2017 - 11 Jun 2017	42	21	59	33	5	21	10	26	5	222
12 Jun 2017 - 25 Jun 2017	63	21	46	16	21	7	23	14	17	228
26 Jun 2017 - 09 Jul 2017	60	18	18	47	17	40	27	68	62	357
10 Jul 2017 - 23 Jul 2017	37	5	46	50	8	8	7	42	11	214
24 Jul 2017 - 06 Aug 2017	47	12	32	47	25	28	29	55	19	294
07 Aug 2017 - 20 Aug 2017	20	17	51	25	28	14	15	18	13	201
21 Aug 2017 - 03 Sep 2017	40	31	16	53	7	30	27	37	45	286
04 Sep 2017 - 17 Sep 2017	36	27	35	14	20	6	8	27	20	193
18 Sep 2017 - 01 Oct 2017	46	12	74	14	5	16	20	24	19	230
02 Oct 2017 - 15 Oct 2017	32	13	56	42	20	14	10	7	10	204
16 Oct 2017 - 29 Oct 2017	11	6	59	42	27	5	15	31	23	219
30 Oct 2017 - 12 Nov 2017	30	12	55	48	27	18	15	13	28	246
13 Nov 2017 - 26 Nov 2017	55	41	50	76	4	39	42	77	24	408
27 Nov 2017 - 10 Dec 2017	26	11	72	29	12	10	6	15	21	202
11 Dec 2017 - 24 Dec 2017	95	55	49	78	7	45	74	108	48	559
25 Dec 2017 - 07 Jan 2018	17	13	33	33	5	16	11	35	21	184
08 Jan 2018 - 21 Jan 2018	18	12	78	55	11	17	11	22	31	255
22 Jan 2018 - 04 Feb 2018	33	32	67	27	4	17	17	19	12	228
05 Feb 2018 - 18 Feb 2018	17	27	71	39	33	10	7	22	21	247
Column Total	1435	1053	2139	1290	623	695	792	1251	881	10159

Table A11: Cross-tabulation of maximum overspending by time period and product category across all households. Entries represent the frequency each category was responsible for the largest increase in irrationality in each time period.

	Dairy	Fruits and vegetables	Meat and Seafood	Non-sugary beverages	Prepared foods	Savory snacks	Starches	Sugary beverages	Sweet snacks	Row Total
25 Jul 2016 - 07 Aug 2016	92	83	114	50	69	63	73	71	41	656
08 Aug 2016 - 21 Aug 2016	83	117	99	59	93	76	88	66	62	743
22 Aug 2016 - 04 Sep 2016	108	104	116	41	68	82	103	67	48	737
05 Sep 2016 - 18 Sep 2016	123	107	108	56	47	70	105	50	60	726
19 Sep 2016 - 02 Oct 2016	67	98	81	52	52	68	78	77	111	684
03 Oct 2016 - 16 Oct 2016	88	113	94	62	59	85	80	60	53	694
17 Oct 2016 - 30 Oct 2016	124	131	144	48	49	98	108	73	73	848
31 Oct 2016 - 13 Nov 2016	80	87	75	53	30	55	69	64	71	584
14 Nov 2016 - 27 Nov 2016	233	195	119	128	49	172	193	218	145	1452
28 Nov 2016 - 11 Dec 2016	102	99	118	49	41	72	106	102	110	799
12 Dec 2016 - 25 Dec 2016	234	219	96	151	58	158	209	211	162	1498
26 Dec 2016 - 08 Jan 2017	120	93	103	97	31	112	114	88	110	868
09 Jan 2017 - 22 Jan 2017	83	61	86	78	63	74	85	82	52	664
23 Jan 2017 - 05 Feb 2017	70	90	75	74	74	79	94	99	61	716
06 Feb 2017 - 19 Feb 2017	71	84	89	47	56	64	83	85	48	627
20 Feb 2017 - 05 Mar 2017	88	94	110	86	44	74	74	92	84	746
06 Mar 2017 - 19 Mar 2017	76	102	112	52	40	50	57	82	56	627
20 Mar 2017 - 02 Apr 2017	93	109	74	54	71	75	78	82	118	754
03 Apr 2017 - 16 Apr 2017	151	123	45	81	60	119	129	134	104	946
17 Apr 2017 - 30 Apr 2017	110	96	115	49	95	103	93	67	79	807
01 May 2017 - 14 May 2017	107	99	99	54	43	83	97	55	55	692
15 May 2017 - 28 May 2017	151	133	107	82	105	100	149	116	48	991
29 May 2017 - 11 Jun 2017	109	75	95	84	30	78	97	100	44	712
12 Jun 2017 - 25 Jun 2017	147	92	89	58	80	81	96	65	78	786
26 Jun 2017 - 09 Jul 2017	187	135	54	122	80	174	176	168	186	1282
10 Jul 2017 - 23 Jul 2017	108	46	89	98	41	77	84	112	59	714
24 Jul 2017 - 06 Aug 2017	160	72	84	114	77	131	137	147	93	1015
07 Aug 2017 - 20 Aug 2017	88	72	104	56	79	77	84	77	58	695
21 Aug 2017 - 03 Sep 2017	136	115	50	122	55	129	125	128	130	990
04 Sep 2017 - 17 Sep 2017	90	85	68	38	64	67	72	105	69	658
18 Sep 2017 - 01 Oct 2017	112	77	124	59	39	85	112	105	54	767
02 Oct 2017 - 15 Oct 2017	90	72	105	85	74	69	71	50	51	667
16 Oct 2017 - 29 Oct 2017	54	62	106	82	68	68	94	86	78	698
30 Oct 2017 - 12 Nov 2017	84	70	122	97	85	85	87	66	99	795
13 Nov 2017 - 26 Nov 2017	198	151	138	170	22	165	211	234	139	1428
27 Nov 2017 - 10 Dec 2017	81	60	119	68	60	67	55	67	71	648
11 Dec 2017 - 24 Dec 2017	283	243	125	176	65	218	282	300	187	1879
25 Dec 2017 - 07 Jan 2018	71	62	61	66	37	76	76	96	84	629
08 Jan 2018 - 21 Jan 2018	92	79	129	108	52	83	81	99	104	827
22 Jan 2018 - 04 Feb 2018	108	106	121	72	26	78	105	102	72	790
05 Feb 2018 - 18 Feb 2018	79	110	121	88	101	90	77	90	84	840
Column Total	4731	4221	4083	3266	2432	3830	4387	4238	3491	34679

Table A12: Cross-tabulation of all overspending by time period and product category across all households. Entries represent the frequency each category contributed to an increase in irrationality in each time period.

Product Category	Display Type	08/06/16 - 07/08/16	08/08/16 - 21/08/16	22/08/16 - 04/09/16	05/09/16 - 18/09/16	19/09/16 - 02/10/16	03/10/16 - 16/10/16	17/10/16 - 30/10/16	31/10/16 - 13/11/16	14/11/16 - 27/11/16	28/11/16 - 11/12/16	12/12/16 - 25/12/16	26/12/16 - 08/01/17	09/01/17 - 22/01/17	23/01/17 - 05/02/17	06/02/17 - 19/02/17	20/02/17 - 05/03/17	06/03/17 - 19/03/17	20/03/17 - 02/04/17	03/04/17 - 16/04/17	17/04/17 - 30/04/17	01/05/17 - 14/05/17	15/05/17 - 28/05/17	29/05/17 - 11/06/17	12/06/17 - 25/06/17	26/06/17 - 09/07/17	10/07/17 - 23/07/17	24/07/17 - 06/08/17	07/08/17 - 20/08/17	21/08/17 - 03/09/17	10/09/17 - 23/09/17	24/09/17 - 07/10/17	08/10/17 - 21/10/17	22/10/17 - 04/11/17	05/11/17 - 18/11/17	Run Total										
Dairy	Front End Cap	129	153	169	168	145	168	117	106	104	154	132	105	137	188	195	127	129	123	100	103	127	121	98	171	139	155	154	133	125	109	111	125	225	229	158	253	319	201	174	220	6271				
	In-Aisle	166	121	212	177	180	136	179	194	262	145	212	214	205	176	194	207	199	196	133	209	246	115	164	225	157	109	163	173	146	275	173	147	239	291	143	7876									
	In-Shelf	38	52	24	21	12	9	14	23	11	10	12	3	5	14	3	10	6	4	14	6	5	18	28	12	7	3	10	6	4	3	10	5	5	13	15	5	18	3	5	13	7	439			
	Mid-Aisle End Cap	0	1	1	4	0	0	0	1	0	2	7	11	10	11	12	16	2	0	0	2	5	6	4	7	8	8	12	4	11	1	7	11	16	22	13	6	10	5	18	5	270				
	Rear End Cap	4	12	10	9	3	4	11	1	6	7	13	15	5	10	12	3	7	12	4	3	29	16	32	19	14	15	4	16	9	6	12	6	4	11	8	16	20	23	15	441					
	Secondary Location	224	196	174	127	159	243	152	142	147	98	110	126	126	131	134	95	118	125	121	128	149	109	132	67	89	88	81	103	142	95	125	133	142	177	131	141	192	75	97	109	5325				
	Side-Aisle End Cap	41	33	41	19	25	17	27	26	23	33	35	15	22	27	33	26	24	49	54	41	29	31	39	20	33	21	52	64	65	43	57	37	45	47	71	31	60	34	70	66	33	1568			
	Store Front	26	24	15	4	18	12	20	12	27	25	16	4	9	10	1	1	11	25	4	0	1	7	3	4	5	22	7	5	5	5	4	19	2	9	1	11	13	14	13	5	430				
	Store Rear	122	68	108	110	106	79	94	117	151	98	186	95	103	130	113	108	96	172	128	112	130	88	135	133	80	155	129	111	101	176	145	130	113	188	103	240	194	127	95	30	5018				
	Subtotal	750	660	754	635	648	608	614	622	733	572	721	568	582	732	651	606	571	855	693	700	581	640	624	571	680	536	833	757	646	557	720	631	663	814	901	606	1043	714	761	874	628	27900			
Fruits and vegetables	Front End Cap	6	7	2	3	4	5	2	5	12	7	10	6	0	2	2	0	0	2	3	1	8	4	2	3	2	2	0	1	0	7	6	1	3	4	4	5	7	163	97	84	28	11	4	2	568
	In-Aisle	10	6	13	7	19	3	15	25	18	16	18	12	11	17	13	10	8	3	14	13	4	18	28	12	7	3	10	6	4	3	10	5	5	13	15	5	18	3	5	13	7	439			
	In-Shelf	37	42	47	67	97	64	85	120	156	130	115	73	82	66	65	83	110	78	94	106	104	74	48	43	29	12	18	4	2	30	83	48	39	86	134	61	48	92	116	80	75	2943			
	Mid-Aisle End Cap	0	1	2	3	1	1	3	6	9	5	8	1	4	4	9	6	2	1	2	0	2	0	3	1	2	1	3	0	0	6	1	0	4	5	4	1	3	2	1	1	108				
	Rear End Cap	14	6	7	16	30	21	28	25	82	42	31	41	40	15	15	7	2	17	19	20	26	9	8	20	7	22	16	15	13	4	23	17	35	30	37	32	35	23	24	20	806				
	Secondary Location	17	7	40	28	16	24	25	6	65	22	42	55	59	70	96	42	60	83	6	10	70	17	22	24	27	43	7	16	9	5	22	37	5	9	17	30	12	17	54	97	1209				
	Side-Aisle End Cap	0	0	0	0	0	1	0	0	6	0	1	1	0	1	1	0	1	2	1	0	1	2	1	2	0	0	0	0	1	4	3	19	10	14	15	22	9	13	26	9	10	355			
	Store Front	3	1	7	5	13	3	4	8	14	15	12	24	13	6	4	7	11	9	5	11	4	14	11	3	10	5	6	1	3	3	19	10	14	15	22	9	13	26	9	10	355				
	Store Rear	2	3	0	1	1	1	1	3	2	2	5	2	1	1	0	4	1	0	4	1	7	3	3	1	1	1	4	3	2	4	1	4	6	1	3	2	9	1	0	1	0	3	96		
	Subtotal	89	73	118	130	181	123	161	198	358	245	241	200	217	179	264	362	140	149	221	168	157	169	118	169	81	74	98	45	45	61	134	110	117	214	407	237	197	194	167	136	6770				
Meat and Seafood	Front End Cap	15	4	8	7	6	13	1	1	2	2	3	103	86	46	52	41	8	37	7	12	6	1	12	6	1	12	4	5	6	4	2	4	5	2	9	2	0	1	1	13	23	569			
	In-Aisle	30	16	12	15	18	19	11	4	15	9	5	21	6	5	26	19	23	14	7	17	28	26	28	33	33	18	14	17	25	15	13	19	8	9	10	13	19	29	45	27	6	823			
	In-Shelf	59	10	48	20	20	9	2	0	3	1	0	24	49	19	14	61	49	41	15	62	51	28	33	17	14	1	51	61	51	25	25	15	5	8	2	2	0	1	44	55	28	20	1891		
	Mid-Aisle End Cap	1	1	3	7	0	5	4	0	0	0	0	1	2	5	7	2	9	10	2	1	4	5	8	10	8	8	16	4	3	1	1	3	4	6	5	10	12	11	195						
	Rear End Cap	38	21	10	18	49	78	11	6	5	16	14	34	20	31	13	45	40	82	37	42	33	14	50	49	25	21	27	26	15	32	23	15	12	12	4	9	11	9	55	36	75	1163			
	Secondary Location	39	52	46	60	68	43	22	25	15	26	33	30	46	81	58	72	107	63	50	73	67	29	63	51	77	49	60	83	43	51	50	27	29	63	51	50	27	29	63	51	50	2076			
	Side-Aisle End Cap	2	9	3	0	4	2	1	2	3	4	2	3	4	0	3	11	1	0	7	26	1	2	2	5	12	4	2	2	6	2	7	2	6	7	2	6	7	2	5	6	180				
	Store Front	13	6	9	7	3	1	1	4	4	2	9	7	3	4	13	12	13	12	8	11	3	9	13	1	5	22	23	16	12	4	17	15	16	12	5	22	5	9	14	33	2	5018			
	Store Rear	273	267	191	200	210	212	211	194	170	219	223	167	212	227	200	191	188	242	213	239	200	215	229	235	233	260	170	195	200	179	175	165	187	169	180	159	171	172	188	153	8222				
	Subtotal	472	386	330	330	387	385	275	222	222	280	282	400	431	419	422	447	364	407	430	426	447	390	421	390	378	395	465	370	381	404	330	291	237	285	238	244	230	299	305	401	407	14751			
Non-sugary beverages	Front End Cap	21	10	6	28	14	16	28	28	23	14	23	4	9	6	16	19	11	27	47	29	24	33	32	33	32	33	18	14	17	25	15	13	19	8	9	10	13	19	29	45	27	6	823		
	In-Aisle	30	43	32	39	29	26	21	21	39	55	50	44	56	41	27	25	42	34	42	38	33	53	29	31	46	47	37	32	24	31	43	57	41	24	22	29	25	29	27	25	1438				
	In-Shelf	8	9	12	9	7	5	3	0	2	1	3	1	1	1	3	1	8	7	9	10	2	1	4	5	8	10	8	8	16	4	3	1	1	7	3	0	6	5	10	12	11	195			
	Mid-Aisle End Cap	7	21	20	14	3	5	8	7	3	7	5	3	7	9	4	0	5	4	1	7	20	15	4	13	15	4	13	9	14	9	5	7	15	19	11	7	11	1	6	10	10	373			
	Rear End Cap	16	24	22	15	20	9	12	7	15	27	13	17	30	17	23	18	26	15	32	15	32	15	16	25																					

Product Category	Display Type	25/07/16 - 07/08/16	08/08/16 - 21/08/16	22/08/16 - 04/09/16	05/09/16 - 18/09/16	19/09/16 - 02/10/16	03/10/16 - 16/10/16	17/10/16 - 30/10/16	31/10/16 - 13/11/16	14/11/16 - 27/11/16	28/11/16 - 11/12/16	12/12/16 - 25/12/16	26/12/16 - 08/01/17	09/01/17 - 22/01/17	23/01/17 - 05/02/17	06/02/17 - 19/02/17	20/02/17 - 05/03/17	06/03/17 - 19/03/17	20/03/17 - 02/04/17	03/04/17 - 16/04/17	17/04/17 - 30/04/17	01/05/17 - 14/05/17	15/05/17 - 28/05/17	29/05/17 - 11/06/17	12/06/17 - 25/06/17	26/06/17 - 09/07/17	10/07/17 - 23/07/17	24/07/17 - 06/08/17	07/08/17 - 20/08/17	21/08/17 - 03/09/17	04/09/17 - 17/09/17	18/09/17 - 01/10/17	02/10/17 - 15/10/17	16/10/17 - 29/10/17	30/10/17 - 12/11/17	13/11/17 - 26/11/17	27/11/17 - 10/12/17	11/12/17 - 24/12/17	25/12/17 - 07/01/18	08/01/18 - 21/01/18	22/01/18 - 04/02/18	05/02/18 - 18/02/18	Row Total																																																																																																																																							
Dairy	Front End Cap	4	9	18	10	3	10	7	19	10	14	3	2	10	16	13	1	5	6	6	2	14	11	9	11	20	13	13	4	8	6	13	4	11	26	5	33	11	8	8	4	410																																																																																																																																								
	In-Aisle	4	4	9	5	6	8	12	30	2	18	9	12	7	2	18	8	5	24	12	11	11	11	15	13	1	29	14	6	4	12	5	2	9	24	6	23	7	4	10	2	423																																																																																																																																								
	In-Shelf	0	0	1	3	1	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18																																																																																																																																					
	Mid-Aisle End Cap	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	13																																																																																																																																				
	Rear End Cap	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15																																																																																																																																			
	Secondary Location	12	7	5	5	4	7	12	6	17	2	9	2	9	5	3	14	4	2	10	10	11	6	4	4	11	0	3	4	13	4	3	9	4	2	18	4	23	2	2	4	2	278																																																																																																																																							
	Side-Aisle End Cap	3	2	2	0	0	1	2	0	0	1	2	2	0	2	1	0	1	1	1	1	1	0	2	3	0	3	1	8	0	1	2	1	2	8	0	3	0	3	1	0	0	61																																																																																																																																							
	Store Front	0	0	0	0	0	0	2	1	6	1	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26																																																																																																																																				
	Store Rear	7	1	12	9	5	3	5	4	15	2	29	1	1	9	1	5	9	8	11	5	8	5	8	10	4	34	3	4	1	17	7	3	2	16	5	20	2	15	5	1	0	313																																																																																																																																							
	Subtotal		30	26	47	36	18	27	44	32	87	18	76	19	24	24	33	47	18	23	52	40	30	43	31	43	53	29	87	36	43	19	40	43	15	29	95	21	112	24	34	29	10	1587																																																																																																																																						
Fruits and vegetables	Front End Cap	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	28																																																																																																																																				
	In-Aisle	0	0	0	1	3	0	2	0	3	0	4	2	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25																																																																																																																																				
	In-Shelf	3	2	0	2	5	4	14	2	13	3	12	4	1	3	6	3	3	5	9	2	5	7	1	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	162																																																																																																																																		
	Mid-Aisle End Cap	0	0	0	2	0	0	2	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9																																																																																																																																			
	Rear End Cap	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	49																																																																																																																																		
	Secondary Location	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	11	57																																																																																																																																	
	Side-Aisle End Cap	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1																																																																																																																																			
	Store Front	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	16																																																																																																																																		
	Store Rear	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6																																																																																																																																			
	Subtotal		5	2	1	6	8	8	20	3	33	9	23	10	4	10	8	4	3	9	17	3	5	15	4	5	3	0	2	0	2	0	9	5	3	4	39	6	28	6	7	9	15	353																																																																																																																																						
Meat and Seafood	Front End Cap	0	0	4	0	0	0	0	0	0	0	0	1	4	3	1	5	1	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	26																																																																																																																																			
	In-Aisle	0	1	2	1	0	0	0	1	0	0	2	0	0	0	0	0	0	0	0	0	1	1	4	3	0	0	1	1	2	0	5	1	2	3	0	2	4	0	4	0	5	0	4	52																																																																																																																																					
	In-Shelf	0	1	0	0	3	0	0	0	0	0	0	0	0	0	3	1	7	4	2	0	6	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	14																																																																																																																																			
	Mid-Aisle End Cap	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7																																																																																																																																			
	Rear End Cap	0	0	0	3	3	4	1	0	0	2	1	4	0	3	0	2	4	6	0	1	3	0	1	7	0	1	1	2	1	1	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	64																																																																																																																																		
	Secondary Location	0	0	1	4	9	0	0	1	1	2	1	2	0	2	4	1	2	4	4	3	1	1	2	1	1	2	1	1	1	1	1	1	1	1	4	0	0	0	0	0	0	0	0	0	0	0	79																																																																																																																																		
	Side-Aisle End Cap	1	2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15																																																																																																																																		
	Store Front	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	24																																																																																																																																	
	Store Rear	18	4	10	8	9	8	16	8	9	12	18	14	9	4	10	7	10	2	10	19	10	9	12	11	7	9	15	10	10	7	9	11	11	9	13	15	11	2	10	6	10	4	412																																																																																																																																						
	Subtotal		20	8	17	16	24	14	20	9	11	18	24	23	17	11	21	20	25	16	17	31	19	14	20	23	9	14	21	21	19	12	16	17	14	13	22	15	17	6	23	17	29	723																																																																																																																																						
Non-sugary beverages	Front End Cap	0	0	1	2	1	0	0	1	3	6	0	2	0	0	0	0	0	1	0	3	0	2	4	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	41																																																																																																																																			
	In-Aisle	0	0	0	0	0	0	0	1	2	4	2	0	0	0	0	1	3	1	3	1	0	1	7	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0</

	Worst				All			
	1[IC > 0]	IC	1[IC > 0]	IC	1[IC > 0]	IC	1[IC > 0]	IC
Store Front	-2.62 (1.12)**	0.00 (0.46)	-9.37 (3.53)***	-2.07 (1.00)**	-0.55 (0.59)	-0.02 (0.16)	-4.83 (1.32)***	-1.08 (0.41)***
Store Rear	-0.33 (0.77)	-0.02 (0.21)	1.55 (1.74)	0.70 (0.58)	-0.42 (0.41)	-0.02 (0.10)	2.57 (0.93)***	0.29 (0.34)
Front End Cap	-0.33 (0.82)	-0.00 (0.25)	6.28 (2.54)**	3.52 (0.83)***	-0.12 (0.41)	-0.02 (0.10)	5.81 (1.44)***	1.32 (0.40)***
Mid-Aisle End Cap	-1.86 (1.18)	-0.08 (0.26)	-3.71 (2.54)	-0.44 (0.57)	-0.44 (0.68)	-0.04 (0.13)	-1.30 (2.31)	-0.12 (0.40)
Rear End Cap	0.93 (1.03)	-0.03 (0.28)	-3.27 (2.36)	-1.71 (0.76)**	0.48 (0.50)	-0.02 (0.12)	-3.62 (1.63)**	-0.61 (0.46)
Side-Aisle End Cap	-0.08 (1.26)	-0.07 (0.32)	-3.07 (2.94)	0.25 (0.61)	-0.26 (0.57)	-0.04 (0.14)	-1.79 (1.22)	-0.13 (0.32)
In-Aisle	0.79 (0.78)	-0.03 (0.23)	1.62 (1.57)	1.30 (0.51)**	0.56 (0.37)	-0.01 (0.09)	1.79 (0.83)**	0.52 (0.18)***
Secondary Location	1.12 (1.15)	-0.06 (0.29)	6.17 (2.98)**	1.77 (0.98)*	0.96 (0.40)**	-0.03 (0.11)	2.50 (1.40)*	0.08 (0.42)
In-Shelf	-0.92 (1.21)	-0.02 (0.28)	-3.85 (2.95)	-0.66 (0.70)	0.05 (0.54)	-0.01 (0.14)	-2.24 (1.57)	-0.26 (0.46)
Fruits and vegetables × Store Front			4.21 (4.23)	0.96 (1.13)			0.02 (1.94)	0.17 (0.54)
Meat and Seafood × Store Front			5.23 (4.33)	0.88 (1.58)			4.18 (2.38)*	1.20 (0.76)
Non-sugary beverages × Store Front			5.42 (4.58)	2.31 (1.29)*			6.77 (2.12)***	1.25 (0.62)**
Prepared foods × Store Front			3.89 (4.16)	1.14 (1.15)			3.51 (2.67)	0.87 (0.51)*
Savory snacks × Store Front			11.94 (5.48)**	3.24 (1.25)***			4.89 (2.54)*	1.19 (0.56)**
Starches × Store Front			5.07 (4.32)	1.17 (1.11)			2.74 (1.81)	0.56 (0.49)
Sugary beverages × Store Front			16.46 (4.28)***	7.73 (1.53)***			11.40 (2.06)***	3.02 (0.84)***
Sweet snacks × Store Front			8.59 (4.56)*	1.23 (1.13)			5.00 (1.98)**	1.34 (0.51)**
Fruits and vegetables × Store Rear			-5.26 (2.70)*	-1.64 (0.70)**			-7.26 (1.50)***	-0.92 (0.44)***
Meat and Seafood × Store Rear			6.43 (3.47)*	3.62 (1.36)***			2.34 (1.69)	1.40 (0.70)**
Non-sugary beverages × Store Rear			-7.89 (2.33)***	-2.96 (0.80)***			-6.37 (1.29)***	-0.76 (0.47)
Prepared foods × Store Rear			-0.90 (2.57)	-0.61 (0.67)			-2.42 (1.57)	-0.31 (0.44)
Savory snacks × Store Rear			-0.06 (2.77)	-1.32 (0.65)**			-3.36 (1.53)**	-0.70 (0.41)*
Starches × Store Rear			-3.39 (2.79)	-1.15 (0.67)*			-5.56 (1.60)***	-1.02 (0.43)**
Sugary beverages × Store Rear			-3.04 (2.90)	-0.89 (0.87)			-1.87 (1.77)	-0.43 (0.56)
Sweet snacks × Store Rear			-2.79 (2.57)	-1.53 (0.71)**			-2.49 (1.64)	-0.11 (0.42)
Fruits and vegetables × Front End Cap			-6.87 (3.63)*	-4.71 (1.11)***			-7.99 (2.04)***	-1.76 (0.57)***
Meat and Seafood × Front End Cap			-12.64 (3.54)***	-5.87 (1.22)***			-11.14 (1.94)***	-2.27 (0.61)***
Non-sugary beverages × Front End Cap			-10.08 (3.25)***	-4.34 (0.98)***			-7.11 (1.75)***	-1.51 (0.49)***
Prepared foods × Front End Cap			-7.52 (3.04)**	-4.21 (0.93)***			-7.47 (1.90)***	-1.63 (0.49)***
Savory snacks × Front End Cap			-4.46 (3.47)	-3.05 (0.92)***			-1.20 (2.09)	-0.91 (0.46)**
Starches × Front End Cap			-8.79 (3.55)**	-3.76 (1.12)***			-8.77 (1.98)***	-2.01 (0.49)***
Sugary beverages × Front End Cap			-4.38 (3.79)	-2.01 (1.31)			-4.74 (2.07)**	-0.70 (0.57)
Sweet snacks × Front End Cap			-4.79 (3.23)	-3.70 (1.00)***			-4.89 (1.97)**	-1.32 (0.56)**
Fruits and vegetables × Mid-Aisle End Cap			-2.64 (3.17)	0.02 (0.71)			-3.47 (3.53)	-0.52 (0.68)
Meat and Seafood × Mid-Aisle End Cap			-0.14 (3.06)	-0.95 (1.21)			-3.24 (2.56)	-1.02 (0.70)
Non-sugary beverages × Mid-Aisle End Cap			11.84 (3.65)***	1.63 (0.95)*			7.54 (2.60)***	0.95 (0.55)*
Prepared foods × Mid-Aisle End Cap			-1.94 (3.43)	-0.25 (0.74)			-1.27 (2.62)	0.17 (0.47)
Savory snacks × Mid-Aisle End Cap			2.79 (3.92)	0.63 (0.77)			5.35 (3.39)	0.18 (0.47)
Starches × Mid-Aisle End Cap			-2.45 (3.35)	-0.04 (0.68)			0.51 (3.21)	0.25 (0.47)
Sugary beverages × Mid-Aisle End Cap			6.93 (6.07)	2.43 (1.59)			0.71 (3.06)	0.59 (0.60)
Sweet snacks × Mid-Aisle End Cap			2.32 (3.64)	-0.20 (0.88)			1.62 (3.14)	0.12 (0.49)
Fruits and vegetables × Rear End Cap			0.98 (3.46)	1.35 (0.87)			2.56 (2.17)	0.63 (0.56)
Meat and Seafood × Rear End Cap			-0.01 (3.51)	1.19 (1.31)			0.64 (2.30)	0.05 (0.73)
Non-sugary beverages × Rear End Cap			6.04 (3.99)	2.11 (0.96)**			5.92 (2.28)***	1.41 (0.60)**
Prepared foods × Rear End Cap			3.62 (3.45)	1.12 (0.89)			3.44 (2.26)	0.08 (0.58)
Savory snacks × Rear End Cap			8.98 (3.51)**	2.59 (1.01)**			9.23 (2.39)***	0.97 (0.55)*
Starches × Rear End Cap			6.43 (4.11)	2.25 (0.92)**			6.16 (2.53)**	0.39 (0.57)
Sugary beverages × Rear End Cap			10.75 (3.25)***	2.84 (1.15)**			5.37 (2.41)**	0.95 (0.65)
Sweet snacks × Rear End Cap			1.07 (3.13)	1.65 (0.96)*			3.57 (2.41)	0.76 (0.60)
Fruits and vegetables × Side-Aisle End Cap			-6.40 (3.83)*	-1.96 (0.93)**			-2.33 (2.15)	0.12 (0.49)
Meat and Seafood × Side-Aisle End Cap			-4.46 (4.15)	-2.94 (1.22)**			-6.19 (1.83)***	-1.57 (0.73)**
Non-sugary beverages × Side-Aisle End Cap			7.01 (4.44)	-0.19 (1.16)			3.18 (2.49)	-0.04 (0.68)
Prepared foods × Side-Aisle End Cap			-0.74 (3.68)	-0.63 (0.75)			-0.11 (2.27)	0.26 (0.51)
Savory snacks × Side-Aisle End Cap			9.37 (4.93)*	1.01 (0.76)			4.91 (2.04)**	-0.22 (0.44)
Starches × Side-Aisle End Cap			8.53 (5.10)*	1.30 (1.14)			9.28 (2.03)***	1.24 (0.48)**
Sugary beverages × Side-Aisle End Cap			7.67 (4.38)*	0.06 (1.19)			4.54 (2.07)**	0.90 (0.63)
Sweet snacks × Side-Aisle End Cap			5.94 (4.34)	0.44 (0.88)			0.46 (1.74)	0.08 (0.59)
Fruits and vegetables × In-Aisle			-4.81 (2.30)**	-2.39 (0.65)***			-5.36 (1.30)***	-1.08 (0.28)***
Meat and Seafood × In-Aisle			-8.97 (2.22)***	-5.19 (0.83)***			-6.60 (1.25)***	-1.93 (0.48)***
Non-sugary beverages × In-Aisle			-6.48 (2.49)***	-3.37 (0.76)***			-2.99 (1.26)**	-0.83 (0.37)**
Prepared foods × In-Aisle			-3.21 (2.28)	-1.91 (0.61)***			-3.52 (1.25)***	-0.91 (0.31)***
Savory snacks × In-Aisle			1.85 (2.37)	-1.36 (0.60)**			-0.63 (1.29)	-0.83 (0.27)***
Starches × In-Aisle			8.18 (2.20)***	0.96 (0.65)			5.79 (1.20)***	0.28 (0.24)
Sugary beverages × In-Aisle			7.40 (2.35)***	2.34 (0.92)**			4.94 (1.21)***	1.05 (0.28)***
Sweet snacks × In-Aisle			-1.47 (2.51)	-1.10 (0.67)*			-2.67 (1.41)*	-0.50 (0.33)
Fruits and vegetables × Secondary Location			-8.12 (4.07)**	-2.05 (1.11)*			-1.61 (1.77)	0.02 (0.50)
Meat and Seafood × Secondary Location			-4.89 (4.00)	-1.98 (1.29)			-4.08 (1.91)**	-0.64 (0.67)
Non-sugary beverages × Secondary Location			-11.06 (3.93)***	-3.50 (1.21)***			-3.23 (1.90)*	-0.27 (0.63)
Prepared foods × Secondary Location			-0.83 (4.41)	-2.07 (1.10)*			0.08 (1.85)	0.22 (0.52)
Savory snacks × Secondary Location			0.55 (3.59)	-0.85 (1.08)			1.62 (2.02)	0.26 (0.50)
Starches × Secondary Location			-2.91 (4.60)	-0.73 (1.19)			0.57 (1.93)	0.18 (0.48)
Sugary beverages × Secondary Location			-5.15 (3.86)	-1.86 (1.54)			-2.56 (1.90)	-0.19 (0.63)
Sweet snacks × Secondary Location			-13.05 (3.77)***	-3.50 (1.18)***			-4.68 (1.74)***	-0.51 (0.52)
Fruits and vegetables × In-Shelf			4.38 (4.53)	1.19 (1.06)			8.05 (2.62)***	0.84 (0.62)
Meat and Seafood × In-Shelf			-4.43 (4.04)	-0.86 (1.15)			0.52 (2.74)	0.14 (0.86)
Non-sugary beverages × In-Shelf			-1.61 (3.95)	0.18 (0.89)			0.66 (2.42)	0.25 (0.69)
Prepared foods × In-Shelf			-0.12 (3.55)	0.37 (0.79)			0.73 (2.17)	-0.06 (0.57)
Savory snacks × In-Shelf			6.02 (4.13)	0.71 (0.83)			2.66 (2.34)	0.22 (0.59)
Starches × In-Shelf			6.45 (3.75)*	0.70 (0.75)			1.90 (2.29)	0.04 (0.52)
Sugary beverages × In-Shelf			6.59 (3.89)*	1.97 (1.13)*			3.36 (2.29)	0.94 (0.65)
Sweet snacks × In-Shelf			9.10 (3.85)**	1.45 (0.86)*			2.75 (2.44)	-0.11 (0.61)
Household FE	✓	✓	✓	✓	✓	✓	✓	✓
Category FE	✓	✓	✓	✓	✓	✓	✓	✓
Worst Period FE	✓	✓	✓	✓	✓	✓	✓	✓
Num. obs.	91431	91431	91431	91431	312111	312111	312111	312111
Num. groups: household	1748	1748	1748	1748	1748	1748	1748	1748
Num. groups: category	9	9	9	9	9	9	9	9
Num. groups: time	41	41	41	41	41	41	41	41
R ² (full model)	0.06	0.00	0.06	0.00	0.05	0.00	0.05	0.00
R ² (proj model)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

***p < 0.01; **p < 0.05; *p < 0.1.

Table A15: Regressions of non-regular display on consumer irrationality (denoted by irrationality contribution (IC)).

	Worst				All			
	1[IC > 0]	IC	1[IC > 0]	IC	1[IC > 0]	IC	1[IC > 0]	IC
Any Display	-0.07 (0.63)	-0.07 (0.19)	2.97** (1.31)	1.88*** (0.49)	0.16 (0.34)	-0.05 (0.09)	2.41*** (0.88)	0.53 (0.33)
Fruits and vegetables $\times$ Any Display			-8.60*** (2.00)	-3.76*** (0.68)			-6.63*** (1.21)	-1.25*** (0.45)
Meat and Seafood $\times$ Any Display			-4.53** (2.19)	-2.46*** (0.78)			-5.50*** (1.39)	-1.13* (0.63)
Non-sugary beverages $\times$ Any Display			-8.95*** (2.11)	-4.00*** (0.87)			-4.74*** (1.26)	-0.98* (0.57)
Prepared foods $\times$ Any Display			-6.08*** (2.09)	-3.06*** (0.72)			-4.48*** (1.23)	-1.01** (0.51)
Savory snacks $\times$ Any Display			2.31 (2.42)	-1.21** (0.61)			2.21 (1.68)	-0.43 (0.43)
Starches $\times$ Any Display			1.14 (2.06)	-0.61 (0.63)			1.54 (1.33)	-0.43 (0.39)
Sugary beverages $\times$ Any Display			2.05 (2.00)	0.29 (0.74)			1.79 (1.23)	0.72 (0.46)
Sweet snacks $\times$ Any Display			-4.65** (1.80)	-2.72*** (0.74)			-4.47*** (1.17)	-0.69 (0.49)
Household FE	✓	✓	✓	✓	✓	✓	✓	✓
Category FE	✓	✓	✓	✓	✓	✓	✓	✓
Worst Period FE	✓	✓	✓	✓	✓	✓	✓	✓
Num. obs.	91431	91431	91431	91431	312111	312111	312111	312111
Num. groups: household	1748	1748	1748	1748	1748	1748	1748	1748
Num. groups: category	9	9	9	9	9	9	9	9
Num. groups: time	41	41	41	41	41	41	41	41
R ² (full model)	0.06	0.00	0.06	0.00	0.05	0.00	0.05	0.00
R ² (proj model)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table A16: Regressions of any form on non-regular display on consumer irrationality (denoted by irrationality contribution (IC)).

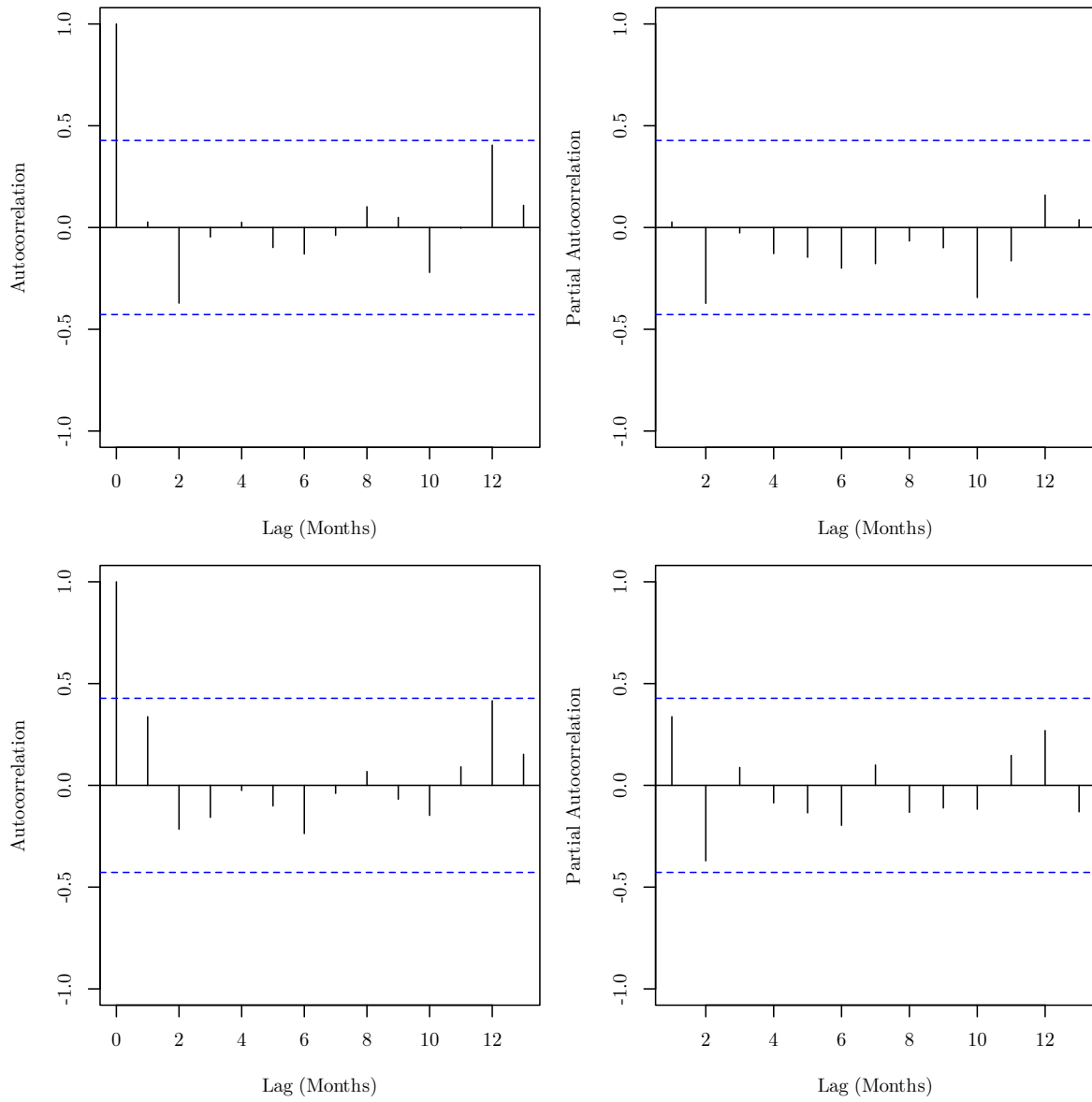
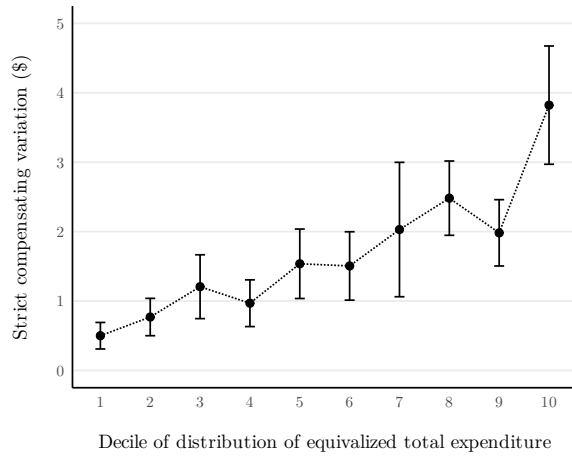


Figure A1: Autocorrelation and Partial Autocorrelation Functions for CE and SS time series, respectively.

(A)



(B)

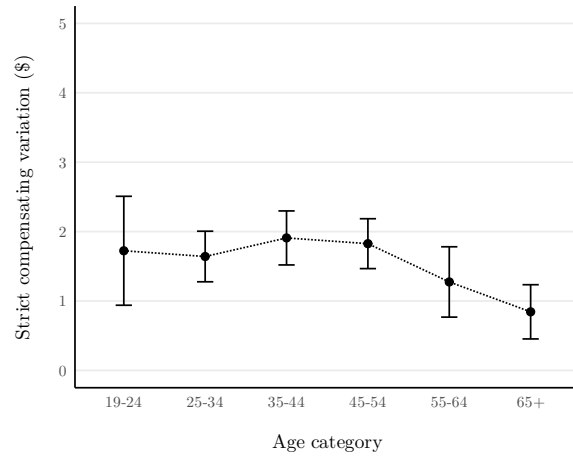


Figure A2: Strict compensating variation by age and expenditure

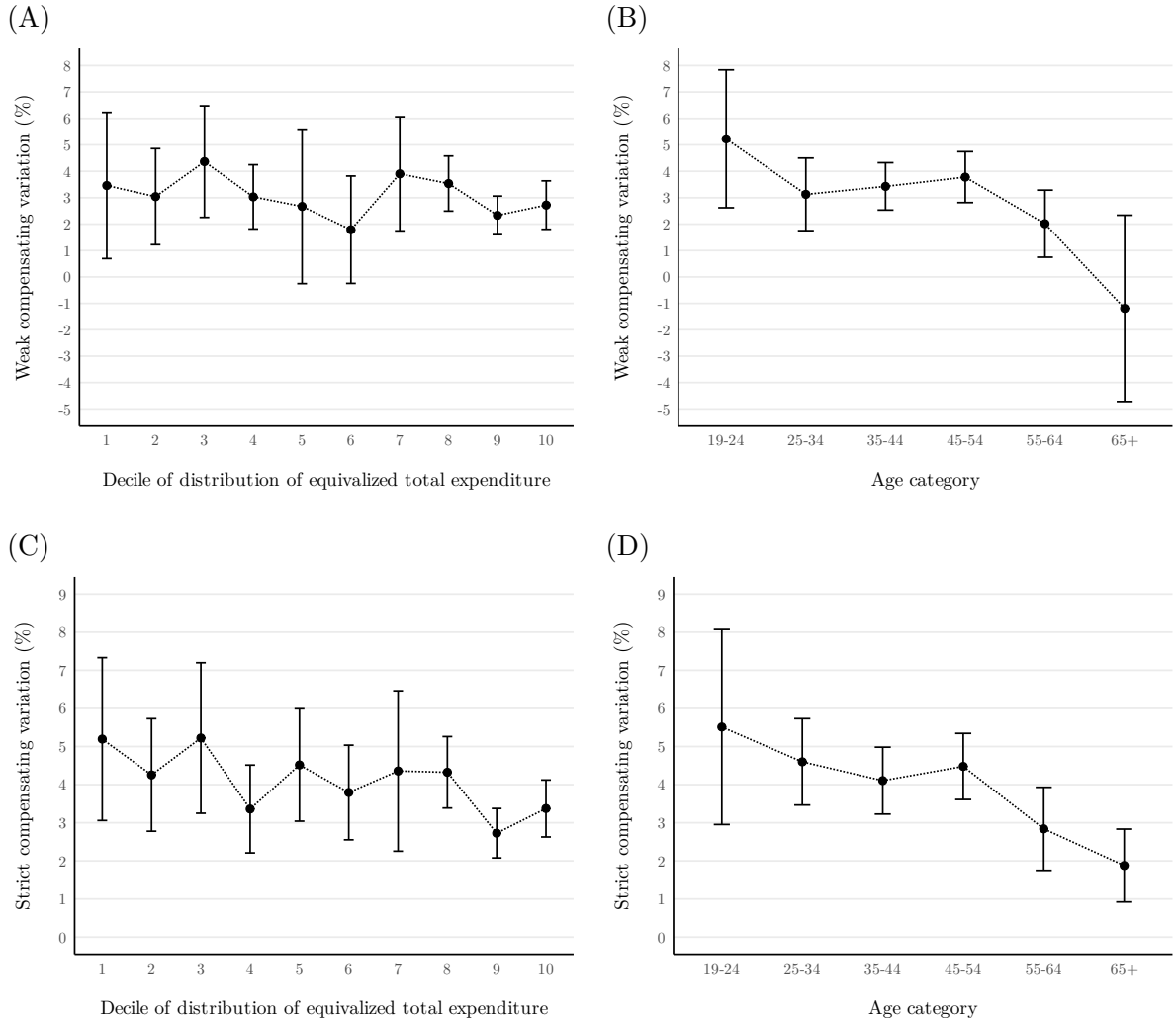


Figure A3: Weak and strict compensating variation (as percentages of equivalized expenditure) by age and expenditure

For Figures A4 and A5 below, to impute income values from the reported income bands, we assume that the underlying household income distribution is log-normal (Blundell, Horowitz, and Parey, 2012), i.e. $\ln(X) \sim N(\mu, \sigma^2)$. We estimate the parameters using a simple Tobit regression so that the specific income assigned to a household reporting an income within the interval $[a, b]$ is calculated as a conditional expectation ($E[X \mid a < X < b]$). For the lowest income bracket starting at zero, we evaluate the limit as $a \rightarrow 0$. Conversely, for the highest income bracket, we evaluate the limit as $b \rightarrow \infty$.

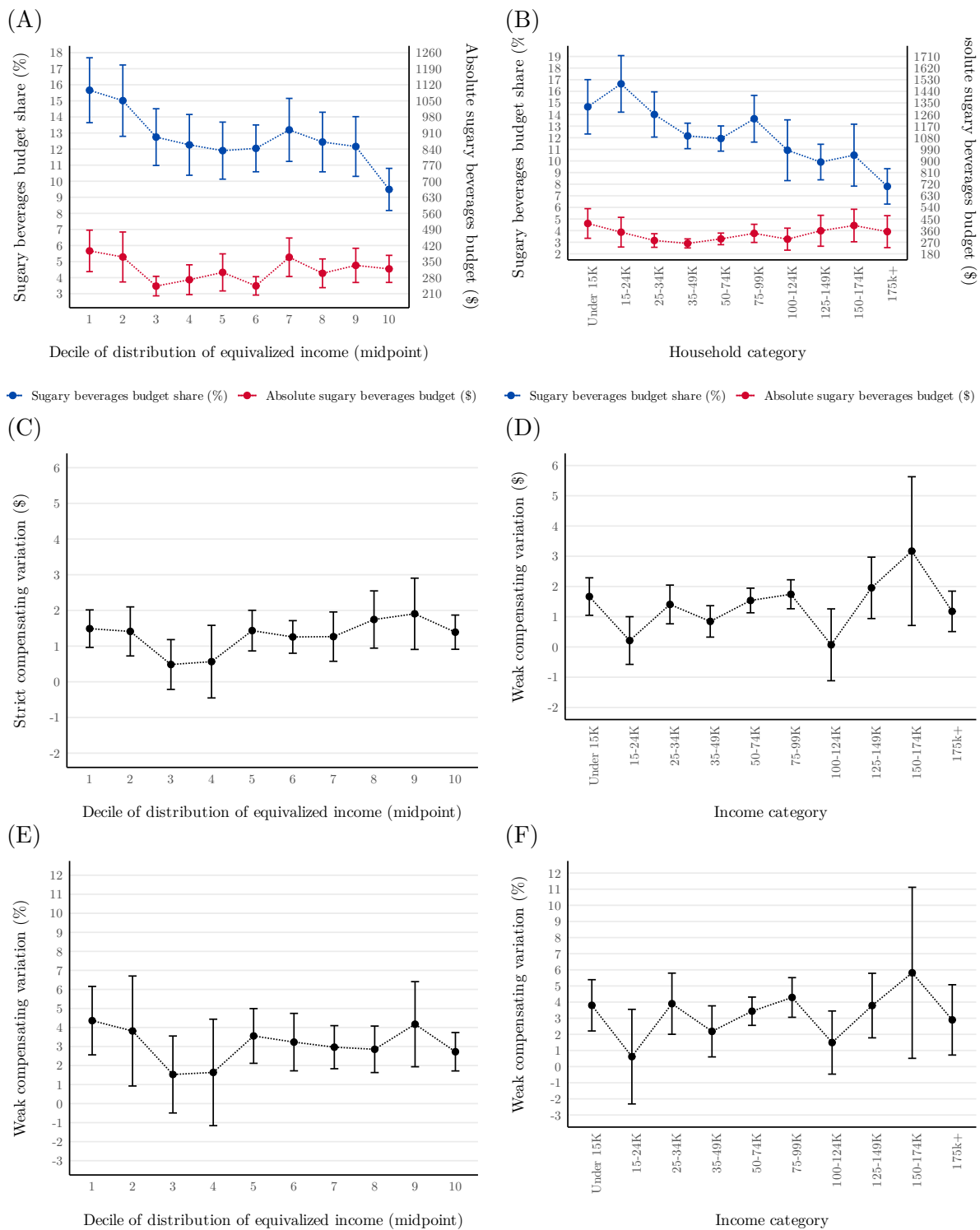


Figure A4: Weak Compensating Variation Analysis Across Different Income Measures

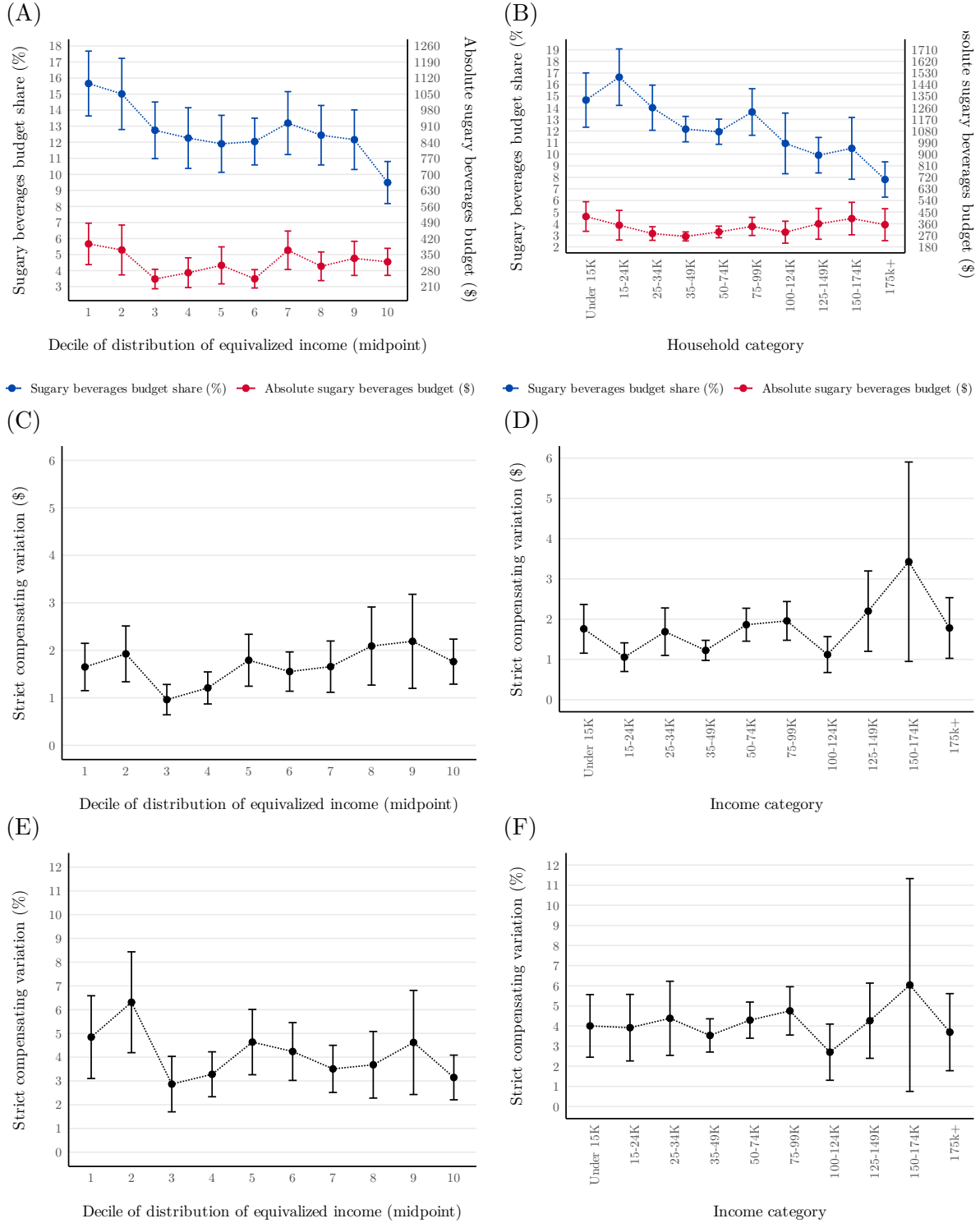


Figure A5: Strict Compensating Variation Analysis Across Different Income Measures

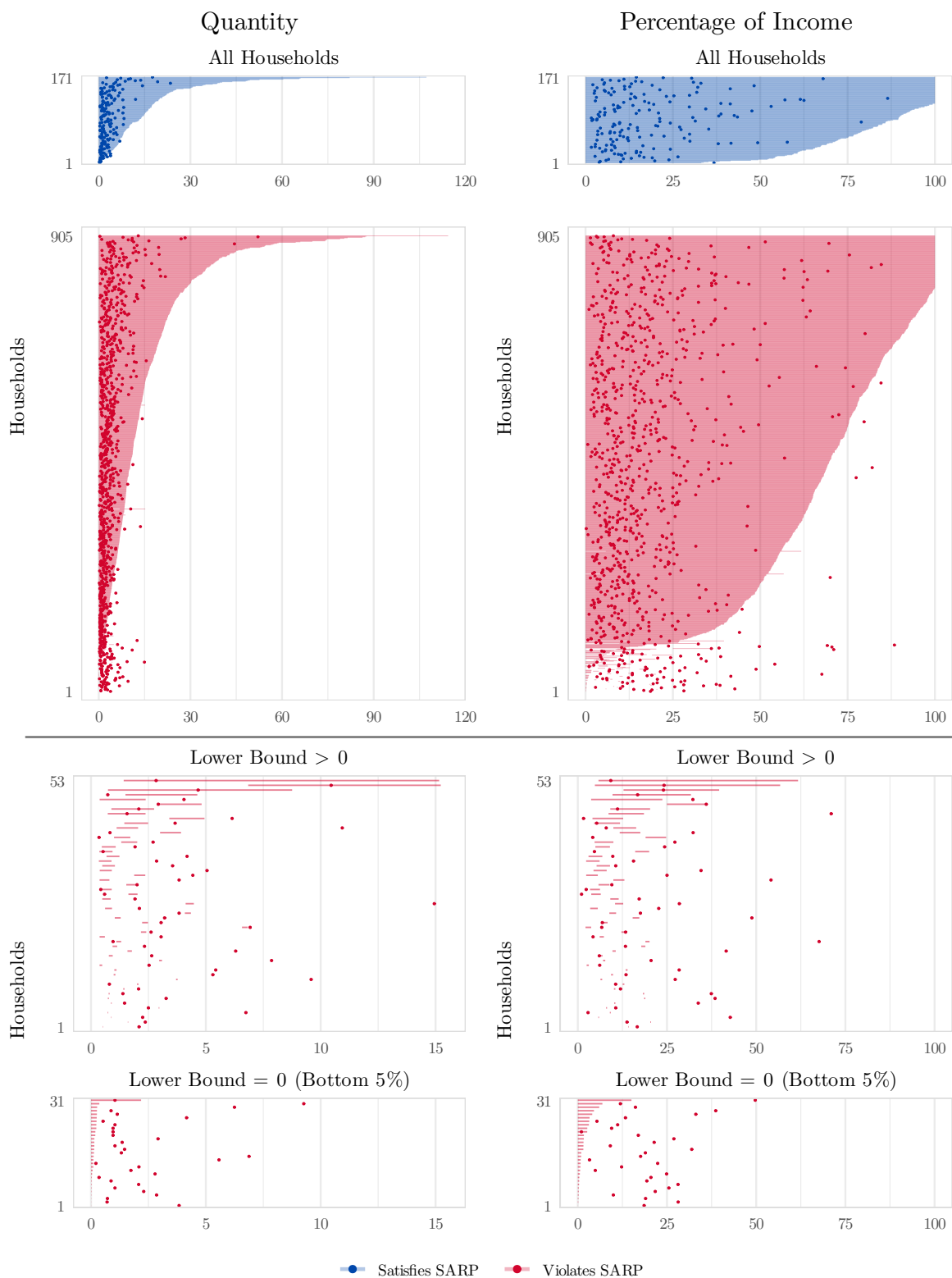


Figure A6: Estimated lower and upper bounds of counterfactual demand for sugary beverages across households.

B Online Appendix

B.1 Sample Construction

1. **Product categorization:** The raw dataset captures transactions across distinct Universal Product Codes (UPCs, indexed by j). Using granular commodity and sub-commodity descriptions, we choose products from grocery departments and exclude non-consumable household supplies (such as pet care) and any storable products. We assign all products to one of 9 refined subcategories (indexed by $c \in \mathcal{C}$): (1) Dairy, (2) Fruits and Vegetables, (3) Meat and Seafood, (4) Non-sugary Beverages, (5) Prepared Foods, (6) Savory Snacks, (7) Starches and Grains, (8) Sugary Beverages, and (9) Sweet Snacks.
2. **Marketing data:** Store-level marketing records (display placement intensity and advertising mailer) are linked to transaction records by UPC j , store ID s , and calendar week t .
3. **Time period allocation:** While the raw transactions in our dataset record daily purchase dates, each transaction is also mapped to a corresponding calendar week t . We aggregate these weekly records into fortnightly (two-week) consumption periods (to balance observations and issues of storability), indexed by τ . We trim initial and terminal periods where aggregate sample transaction volume falls below 95% of the median period transaction frequency. We sequentially re-index the remaining periods starting from $\tau = 1$.
4. **Price index construction:** Following the Dunnhumby documentation, prices are defined as:

$$p_{ijt} = \frac{\text{Sales Value}_{ijt} - \text{Coupon Discount}_{ijt}}{\text{Quantity}_{ijt}}$$

Within each fortnight τ , we aggregate expenditures and quantities for each household i and UPC j . Total quantity $q_{ij\tau}$ and total expenditure $e_{ij\tau}$ are given by:

$$q_{ij\tau} = \sum_{t \in \tau} \text{Quantity}_{ijt}, \quad e_{ij\tau} = \sum_{t \in \tau} \text{Quantity}_{ijt} \cdot p_{ijt}$$

The corresponding household-specific average price for UPC j during period τ is computed as the mean price across transactions:

$$\tilde{p}_{ij\tau} = \frac{1}{N_{ij\tau}} \sum_{t \in \tau} p_{ijt}$$

where $N_{ij\tau}$ is the number of purchase transactions for product j by household i during period τ . For each UPC $j \in c$, let $E_{j\tau} = \sum_i q_{ij\tau} \tilde{p}_{ij\tau}$ represent aggregate market expenditure across purchasing households, and let $\bar{p}_{j\tau} = \frac{1}{H_{j\tau}} \sum_i \tilde{p}_{ij\tau}$ denote the average product price (where $H_{j\tau}$ is the total number of purchasing households). The market budget share of UPC j within category c during period τ is defined as:

$$s_{j\tau} = \frac{E_{j\tau}}{\sum_{k \in c} E_{k\tau}}$$

The representative market price index for category c in period τ , denoted $P_{c\tau}$, is calculated as the expenditure-share-weighted average price for each category:

$$P_{c\tau} = \sum_{j \in c} s_{j\tau} \bar{p}_{j\tau}$$

Any subcategory exhibiting a zero average price or missing transaction records across the aggregate consumer base in any time period is excluded.

5. **Household quantities construction:** For each household i , category c , and fortnight period τ , total household category expenditure is obtained by summing over all purchased UPCs:

$$E_{ic\tau} = \sum_{j \in c} e_{ij\tau}$$

We compute the aggregate consumption quantity $Q_{ic\tau}$ for household i in category c during period τ by dividing observed total expenditure by the respective category price:

$$Q_{ic\tau} = \frac{E_{ic\tau}}{P_{c\tau}}$$

Finally, we check that all 9 categories are purchased by at least 85% of sample households across all periods. We also keep households that record non-zero consumption in at least two time periods and that consume across at least two distinct categories over the sample duration. This leaves us with a balanced panel of 2,474 households observed across 41 fortnightly periods over 9 categories.

B.2 Time series

References

- Alberto Abadie, Susan Athey, Guido W Imbens, and Jeffrey M Wooldridge. When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35, 02 2023. ISSN 0033-5533. doi: 10.1093/qje/qjac038. URL <https://doi.org/10.1093/qje/qjac038>.
- Sydney Afriat. The construction of utility functions from expenditure data. *International Economic Review*, 8(1):67–77, 1967. ISSN 00206598, 14682354. URL <http://www.jstor.org/stable/2525382>.
- Mark Aguiar and Erik Hurst. Life-cycle prices and production. *American Economic Review*, 97(5):1533–1559, December 2007. doi: 10.1257/aer.97.5.1533. URL <https://www.aeaweb.org/articles?id=10.1257/aer.97.5.1533>.
- Hunt Allcott, Benjamin B. Lockwood, and Dmitry Taubinsky. Should We Tax Sugar-Sweetened Beverages? An Overview of Theory and Evidence. *Journal of Economic Perspectives*, 33(3):202–227, August 2019a. ISSN 0895-3309. doi: 10.1257/jep.33.3.202. URL <https://www.aeaweb.org/articles?id=10.1257%2Fjep.33.3.202>.
- Hunt Allcott, Benjamin B Lockwood, and Dmitry Taubinsky. Regressive Sin Taxes, with an Application to the Optimal Soda Tax*. *The Quarterly Journal of Economics*, 134(3):1557–1626, August 2019b. ISSN 0033-5533. doi: 10.1093/qje/qjz017. URL <https://doi.org/10.1093/qje/qjz017>.
- Jose Apesteguia and Miguel A. Ballester. A measure of rationality and welfare. *Journal of Political Economy*, 123(6):1278–1310, 2015. ISSN 00223808, 1537534X. URL <https://www.jstor.org/stable/10.1086/683838>.
- Manuel Arellano and Stephen Bond. Some tests of specification for panel data: Monte carlo evidence and an application to employment equations. *The Review of Economic Studies*, 58(2):277–297, 04 1991. ISSN 0034-6527. doi: 10.2307/2297968. URL <https://doi.org/10.2307/2297968>.
- Richard Blundell and Stephen Bond. Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87(1):115–143, 1998. ISSN 0304-4076. doi: [https://doi.org/10.1016/S0304-4076\(98\)00009-8](https://doi.org/10.1016/S0304-4076(98)00009-8). URL <https://www.sciencedirect.com/science/article/pii/S0304407698000098>.
- Richard Blundell, Joel L. Horowitz, and Matthias Parey. Measuring the price responsiveness of gasoline demand: Economic shape restrictions and nonparametric demand

- estimation. *Quantitative Economics*, 3(1):29–51, 2012. doi: <https://doi.org/10.3982/QE91>. URL <https://onlinelibrary.wiley.com/doi/abs/10.3982/QE91>.
- Richard Blundell, Martin Browning, Laurens Cherchye, Ian Crawford, Bram De Rock, and Frederic Vermeulen. Sharp for SARP: Nonparametric bounds on the behavioural and welfare effects of price changes. *American Economic Journal: Microeconomics*, 7(1):43–60, 2015.
- Richard W. Blundell, Martin Browning, and Ian A. Crawford. Nonparametric engel curves and revealed preference. *Econometrica*, 71(1):205–240, 2003. doi: <https://doi.org/10.1111/1468-0262.00394>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0262.00394>.
- Richard W. Blundell, Martin Browning, and Ian A. Crawford. Best nonparametric bounds on demand responses. *Econometrica*, 76(6):1227–1262, 2008.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Robust inference with multiway clustering. *Journal of Business and Economic Statistics*, 29(2):238–249, 2011. ISSN 07350015. URL <http://www.jstor.org/stable/25800796>.
- Pierre Chandon, J. Wesley Hutchinson, Eric T. Bradlow, and Scott H. Young. Does in-store marketing work? effects of the number and position of shelf facings on brand attention and evaluation at the point of purchase. *Journal of Marketing*, 73(6):1–17, 2009. doi: 10.1509/jmkg.73.6.1. URL <https://doi.org/10.1509/jmkg.73.6.1>.
- Mark Dean and Daniel Martin. Measuring Rationality with the Minimum Cost of Revealed Preference Violations. *The Review of Economics and Statistics*, 98(3):524–534, 07 2016. ISSN 0034-6535. doi: 10.1162/REST_a_00542. URL https://doi.org/10.1162/REST_a_00542.
- Thomas Demuynck and John Rehbeck. Computing revealed preference goodness-of-fit measures with integer programming. *Economic Theory*, 2023.
- Sanjay K. Dhar, Stephen J. Hoch, and Nanda Kumar. Effective category management depends on the role of the category. *Journal of Retailing*, 77(2):165–184, 2001. ISSN 0022-4359. doi: [https://doi.org/10.1016/S0022-4359\(01\)00045-8](https://doi.org/10.1016/S0022-4359(01)00045-8). URL <https://www.sciencedirect.com/science/article/pii/S0022435901000458>.
- Pierre Dubois, Rachel Griffith, and Martin O’Connell. How well targeted are soda taxes? *American Economic Review*, 110(11):3661–3704, November 2020. doi: 10.1257/aer.20171898. URL <https://www.aeaweb.org/articles?id=10.1257/aer.20171898>.

- Federico Echenique, Sangmok Lee, and Matthew Shum. The money pump as a measure of revealed preference violations. *Journal of Political Economy*, 119(6):1201–1223, 2011. ISSN 00223808, 1537534X. URL <http://www.jstor.org/stable/10.1086/665011>.
- Jason M. Fletcher, David E. Frisvold, and Nathan Tefft. Non-Linear Effects of Soda Taxes on Consumption and Weight Outcomes. *Health Economics*, 24(5):566–582, 2015. ISSN 1099-1050. doi: 10.1002/hec.3045. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/hec.3045>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/hec.3045>.
- Yoonju Han, Sandeep R. Chandukala, and Shibo Li. Impact of different types of in-store displays on consumer purchase behavior. *Journal of Retailing*, 98(3):432–452, 2022. ISSN 0022-4359. doi: <https://doi.org/10.1016/j.jretai.2021.10.002>. URL <https://www.sciencedirect.com/science/article/pii/S0022435921000634>.
- M. Houtman and J. A. H. Maks. Determining all maximal data subsets consistent with revealed preference. *Kwantitatieve Methoden*, 6(19):89–104, 1985.
- Rob J. Hyndman and Yeasmin Khandakar. Automatic time series forecasting: The forecast package for r. *Journal of Statistical Software*, 27(3):1–22, 2008. doi: 10.18637/jss.v027.i03. URL <https://www.jstatsoft.org/index.php/jss/article/view/v027i03>.
- J. Jeffrey Inman, Russell S. Winer, and Rosellina Ferraro. The interplay among category characteristics, customer characteristics, and customer activities on in-store decision making. *Journal of Marketing*, 73(5):19–29, 2009. doi: 10.1509/jmkg.73.5.19. URL <https://doi.org/10.1509/jmkg.73.5.19>.
- Andrew Levin, Chien-Fu Lin, and Chia-Shang Chu. Unit root tests in panel data: asymptotic and finite-sample properties. *Journal of Econometrics*, 108(1):1–24, 2002. ISSN 0304-4076. doi: [https://doi.org/10.1016/S0304-4076\(01\)00098-7](https://doi.org/10.1016/S0304-4076(01)00098-7). URL <https://www.sciencedirect.com/science/article/pii/S0304407601000987>.
- G. S. Maddala and Shaowen Wu. A comparative study of unit root tests with panel data and a new simple test. *Oxford Bulletin of Economics and Statistics*, 61(S1): 631–652, 1999. doi: <https://doi.org/10.1111/1468-0084.0610s1631>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0084.0610s1631>.
- OECD. *OECD Framework for Statistics on the Distribution of Household Income, Consumption and Wealth*. OECD, June 2013. ISBN 978-92-64-19482-3 978-92-64-19483-0.

doi: 10.1787/9789264194830-en. URL <https://www.oecd.org/en/publications/framework-for-statistics-on-the-distribution-of-household-income-consumption-and-welfare-9789264194830-en.html>.

David Roodman. A note on the theme of too many instruments. *Oxford Bulletin of Economics and Statistics*, 71(1):135–158, 2009. doi: <https://doi.org/10.1111/j.1468-0084.2008.00542.x>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0084.2008.00542.x>.

Elisa B. Schweiger, Carl-Philip Ahlbom, Jens Nordfält, Anne L. Roggeveen, and Dhruv Grewal. In-store endcap projections and their effect on sales. *Journal of Retailing*, 99(1):5–16, 2023. ISSN 0022-4359. doi: <https://doi.org/10.1016/j.jretai.2022.06.004>. URL <https://www.sciencedirect.com/science/article/pii/S0022435922000446>.

Jing Shen, Junjie Wang, Fan Yang, and Ruopeng An. Impact of soda tax on beverage price, sale, purchase, and consumption in the US: a systematic review and meta-analysis of natural experiments. *Frontiers in Public Health*, 11:1126569, September 2023. ISSN 2296-2565. doi: 10.3389/fpubh.2023.1126569. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10556476/>.

Bart Smeulders, Laurens Cherchye, Frits C. R. Spieksma, and Bram De Rock. The money pump as a measure of revealed preference violations: A comment. *Journal of Political Economy*, 121(6):1248–1258, 2013. ISSN 00223808, 1537534X. URL <http://www.jstor.org/stable/10.1086/674077>.

Pei Jie Tan, Armando Corsi, Justin Cohen, Anne Sharp, Larry Lockshin, William Caruso, and Svetlana Bogomolova. Assessing the sales effectiveness of differently located endcaps in a supermarket. *Journal of Retailing and Consumer Services*, 43:200–208, 2018. ISSN 0969-6989. doi: <https://doi.org/10.1016/j.jretconser.2018.03.015>. URL <https://www.sciencedirect.com/science/article/pii/S0969698917307257>.

Hal R. Varian. The non-parametric approach to demand analysis. *Econometrica*, 50(4): 945–973, July 1982.

Hal R. Varian. Goodness-of-fit in optimizing models. *Journal of Econometrics*, 46(1):125–140, 1990. ISSN 0304-4076. doi: [https://doi.org/10.1016/0304-4076\(90\)90051-T](https://doi.org/10.1016/0304-4076(90)90051-T). URL <https://www.sciencedirect.com/science/article/pii/030440769090051T>.